

Geographical skills

This chapter will look at a range of skills that geographers should be able to use. These include:

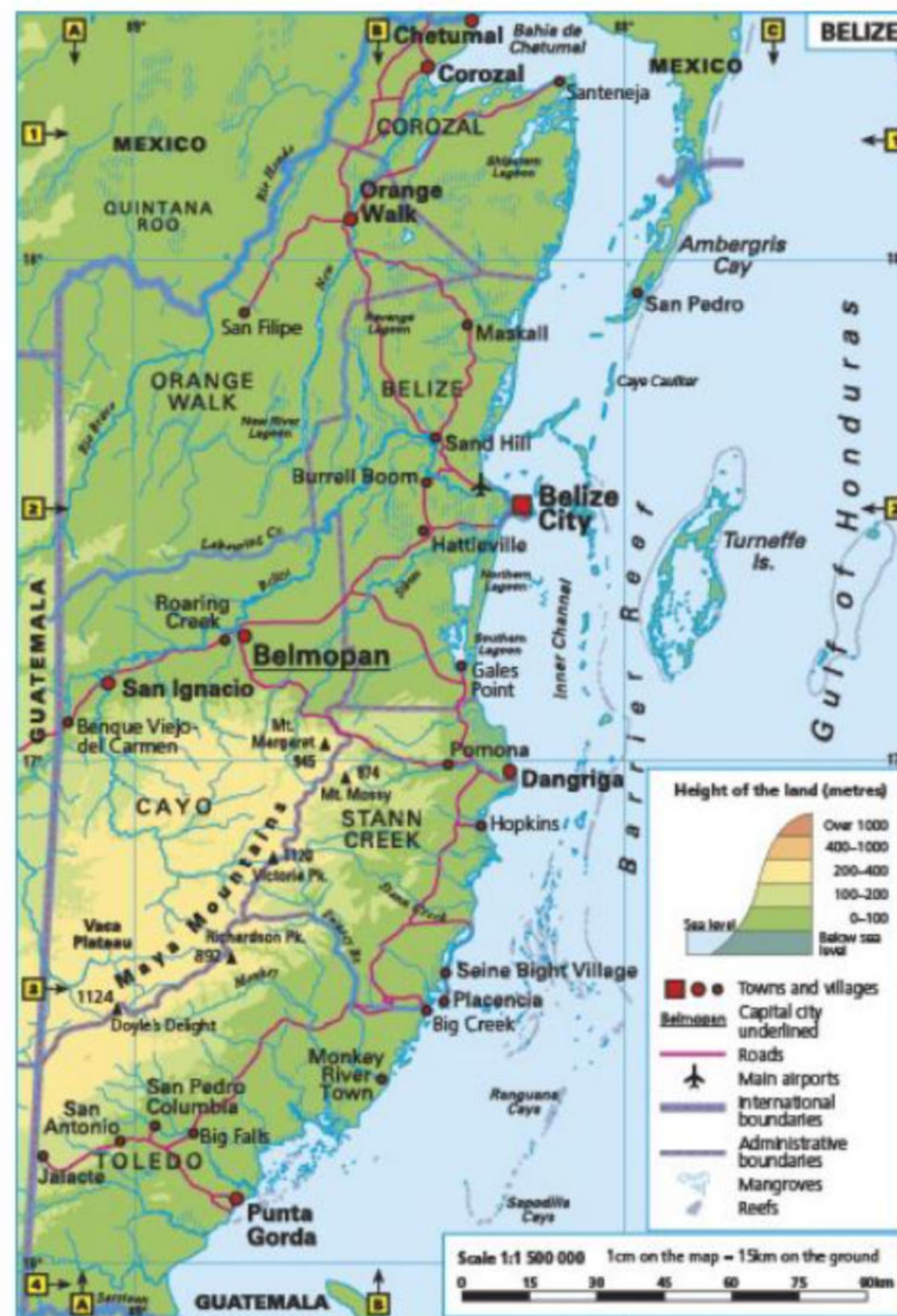
- ★ mapwork (cartographic) skills, such as grid and square references, distance, direction and land use
- ★ graphical skills, such as scatter graphs, pie charts and triangular graphs
- ★ mathematical skills, such as means, ranges and inverse proportions
- ★ fieldwork skills, such as sampling, pilot studies and questionnaires.
- ★ GIS and image skills

Cartographic skills

Atlas maps

Atlas maps come in a variety of **scales**. For example:

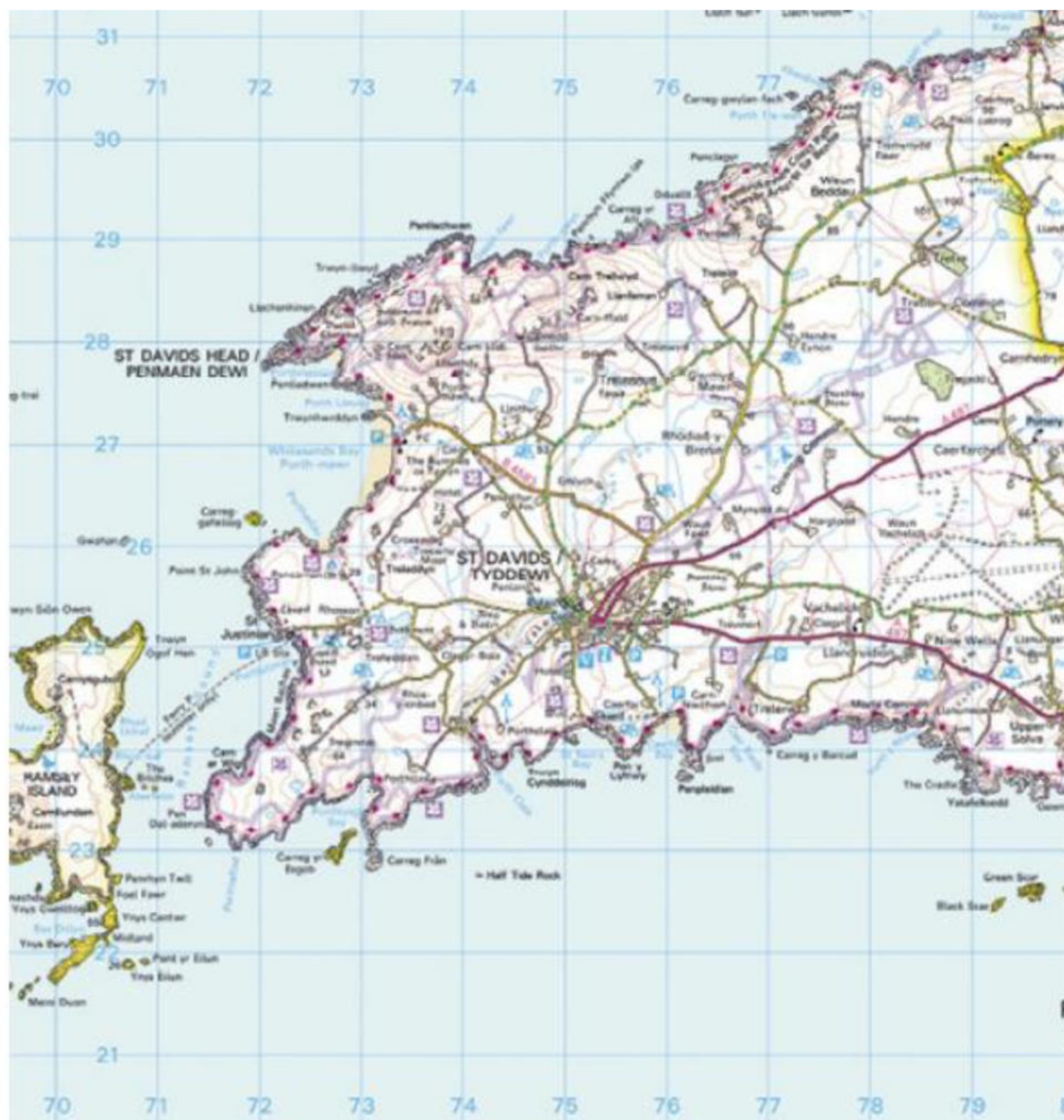
- » A double-page map of southern Britain may be shown at a scale of 1:1,200,000, where 1 cm represents 12 km.
- » A full-page map of the UK may be at a scale of 1:4,000,000. Here, 1 cm is equivalent to 40 km.
- » A map of Europe over two pages might be at a scale of 1:16,000,000, with 1 cm representing 160 km.
- » A map of the entire world set over a double page could be at a scale of 1:77,500,000, so that 1 cm equals 775 km.



▲ **Figure 11.1** An atlas map of Belize

These examples were all taken from a standard school atlas. The scale of an atlas map is generally shown in the lower left-hand or right-hand corner of the page, but this can vary. It is customary for the scale to be given numerically and in a linear format. [Figure 11.1](#) shows an atlas map of Belize.

An Ordnance Survey map extract of part of Wales in the UK, focusing on the city of St Davids, is shown in [Figure 11.2](#).



▲ **Figure 11.2** A 1:50,000 map extract of St Davids, Wales © Crown copyright and database rights 2025 Ordnance Survey (100036470). For Educational Use only

Latitude refers to the angular distance of a location north or south of the equator. In contrast, **longitude** refers to the angular location east or west of Greenwich, UK. On [Figure 11.1](#) the latitudes 16°N, 17°N and 18°N are identified, with the longitudes 88°W and 89°W shown at the top of the map.

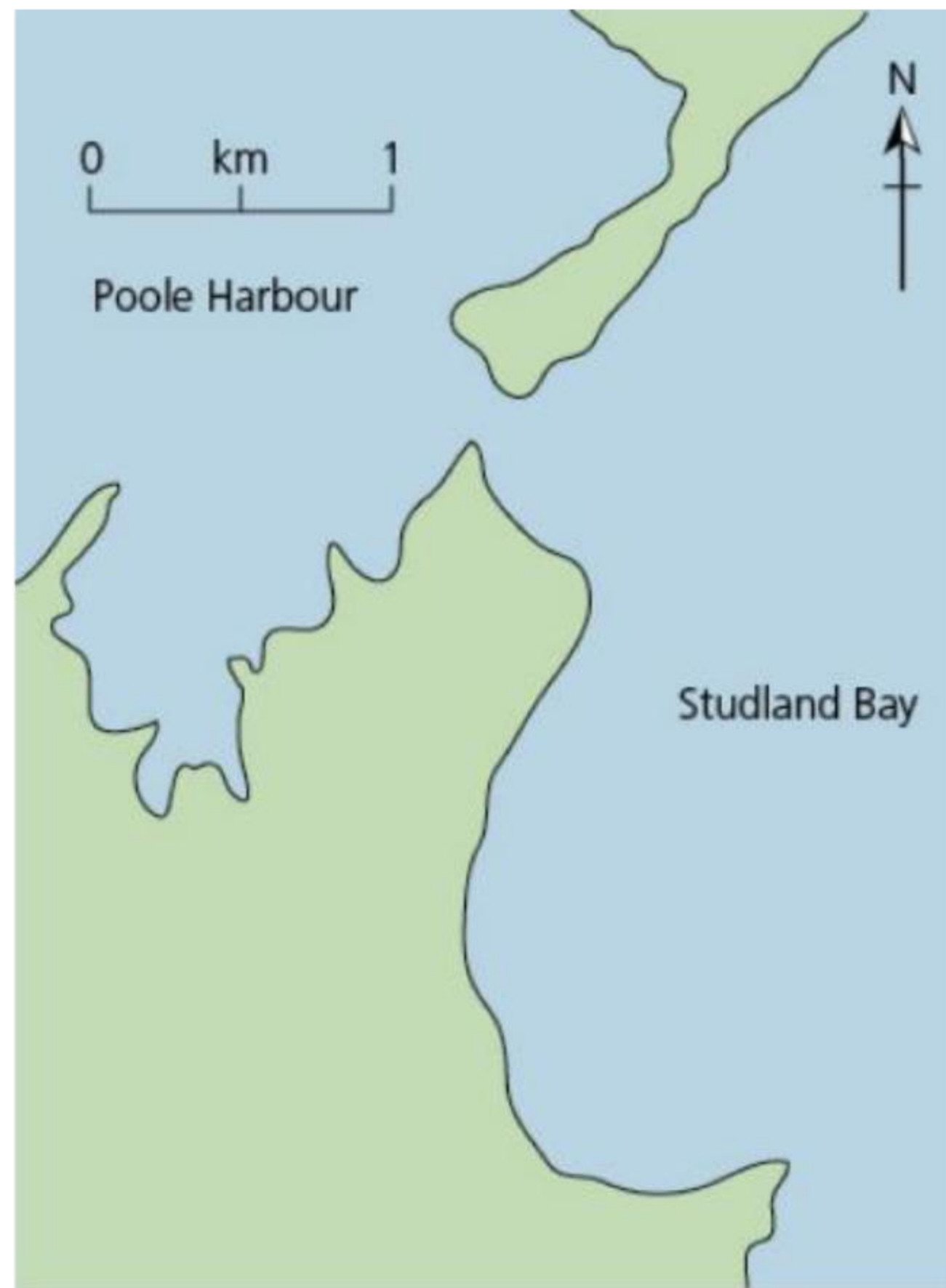
Lines of latitude and longitude are divided into degrees, minutes and seconds. Each degree can be divided into 60 minutes, and each minute into 60 seconds. In most cases only degrees and minutes are used, for example Kingston, Jamaica has a latitude of 18° 0 minutes and London has a latitude of 51° 30 minutes.

Activities

- 1 Look at [Figure 11.1 \(page 291\)](#).
 - a State the direction of Belmopan from Belmopan.
 - b Approximately how far is Belmopan from Belize City?
 - c State the altitude of the Maya Mountains.
- 2 Look at [Figure 11.2 \(page 291\)](#). Estimate the size of the city of St Davids. (All grid squares on Ordnance Survey maps are 1 km by 1 km.)

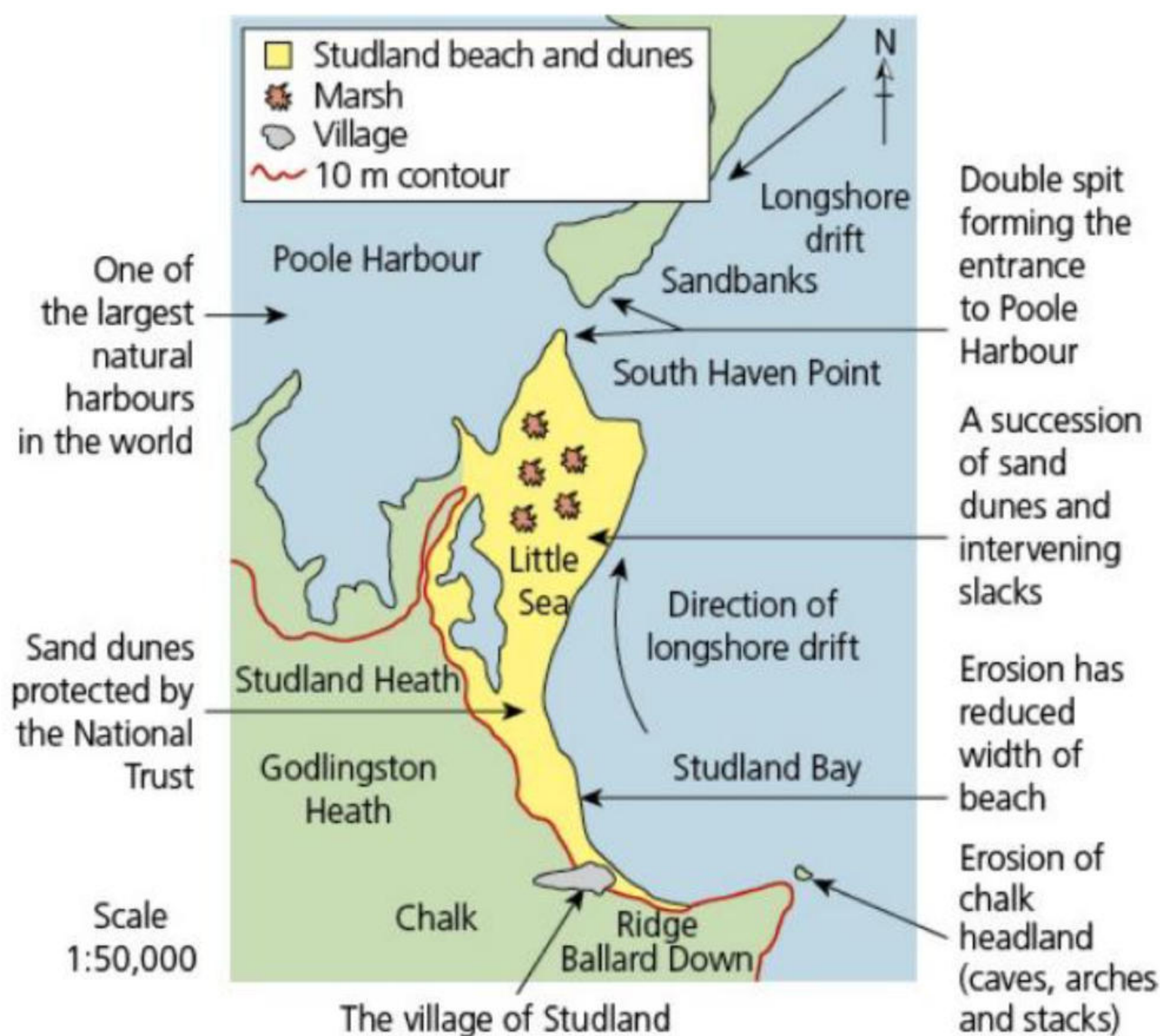
Base maps

A **base map** at the simplest level may be merely an outline of a geographical area, showing just enough information to allow the reader to recognise the area concerned ([Figure 11.3](#)). Because of this, base maps are sometimes referred to as outline maps. In this course, you might have been given outline maps of the world, the UK and other countries and asked to add information to these maps. In investigative work, you might have used base maps that cover a relatively small geographical area, such as a stretch of coastline, a section of a river valley or the central business district of an urban area.



▲ **Figure 11.3** A base map of Studland Beach and Bay

For example, a base map of a coastal area would show the coastline and a few other features to show the limits of the map in each direction. This allows you to be clear about the extent of the geographical area covered by the base map. Information can then be added on a range of other features, such as sand dunes, surface drainage, communications and settlements ([Figure 11.4](#)). Alternatively, a base map may come from a commercial source such as Multimap. Here, more information is presented, but there is still scope for annotation. Thus, you can either draw your own base map or search the internet for the most appropriate base map for the task in hand.



▲ **Figure 11.4** A complete sketch map of Studland Beach and dunes

Photocopied base maps or maps downloaded from the internet need careful adaptation to give them your personal 'stamp'. It is likely that you will want to insert and label a number of features on a base map. **Annotation** takes this process a significant step further by the use of short sentences to add description and maybe some very brief explanation applied to a particular feature or features on a map, diagram or photograph. The objective is to produce an effective piece of geographical communication. The box below is an example of the increasing sequence of sophistication relating to sand dune succession. However,

don't forget to include the basic requirements such as a title, scale and key.

The sequence of annotation

Label: Sand dunes

Descriptive annotation: A succession of sand dunes

Descriptive and explanatory annotation: A succession of sand dunes with newer dunes formed by wind transport and deposition by the coastline.

In this example, the difference between the label and the descriptive annotation is only a matter of three words, but it adds important information. Likewise, the difference between the descriptive and explanatory annotations is the introduction of the two major processes of wind transport and deposition. These are important elements of explanation.

Every illustration you use should have a clear purpose. It should enhance the text and may also have the benefit of freeing up words if a word limit has been placed on the project or investigation you are carrying out. Refer clearly to illustrations in the text by giving them figure numbers. If it is the first illustration to be used, it would be 'Figure 1'. You should briefly comment on what the illustration shows. Base maps often provide the foundation for many or all of the other illustrations that are to follow. Base maps can be used with photographs. For example, a base map showing the course of a river might include small photographs of a waterfall and a meander.

Activities

- 1 **a** Draw a base map of your school. Show the different buildings, along with the different outdoor areas on the school site.
 - b** Provide a label (name) for each building and outdoor area.
 - c** For any four areas on the school site, add an annotation to provide extra information.
 - 2 Look at the box above. Produce another sequence of annotation, beginning with the label 'Vegetation on the dunes'.
-

The features of maps

Scale

Most Ordnance Survey maps that we use are either at a 1:50,000 or a 1:25,000 scale. On a 1:50,000 map, 1 cm on the map relates to 50,000 cm on the ground. On a 1:25,000 map, every 1 cm on the map relates to 25,000 cm on the ground. In every kilometre there are 100,000 centimetres (1000 × 100 cm). This means that:

- » on a 1:50,000 map, every 2 cm corresponds to a kilometre
- » on a 1:25,000 map, every 4 cm corresponds to a kilometre.

A 1:25,000 map is more detailed than a 1:50,000 map and is therefore an excellent source for geographical enquiries. By contrast, a 1:50,000 map provides a more general overview of a larger area. You may also come across other scales, for example 1:10,000 and 1:2500.

Estimating area

To estimate area, count the number of squares. Each square on a 1:50,000 and 1:25,000 map is 1 km². To be more accurate, estimate the percentage of each square that makes up the feature you are measuring, for example the lake in [Figure 11.29 on page 307](#).

Grid and square references

Measurement on maps is made easier by grid lines. These are the regular vertical and horizontal lines you can see on an Ordnance Survey map. The vertical lines are called eastings and the horizontal lines are called northings. They help pinpoint the exact location of features on a map.

Grid references are the **six-figure references** that locate precise positions on a map.

- » The first three figures are the eastings and these tell us how far a position is across the map.
- » The last three figures are the northings and these tell us how far up the map a position is.

An easy way to remember which way round the numbers go is by using the phrase 'along the corridor and up the stairs'. In [Figure 11.5](#), the church at Rose Hill is located at 691045 and the church at Davis Town is found at 737043.

Sometimes a feature covers an area rather than a point, for example the villages and areas of woodland in [Figure 11.5](#). A grid reference for these would be inappropriate, so we use **four-figure square references**.

- » The first two numbers refer to the eastings.
- » The last two numbers refer to the northings.

The point where the two grid lines meet is at the bottom left-hand corner of the square. So in [Figure 11.5](#), most of the village of Seafeld is found in 7504. Some features may occur in two or more squares, for example Long Bay is found both in square 7006 and in 7106.

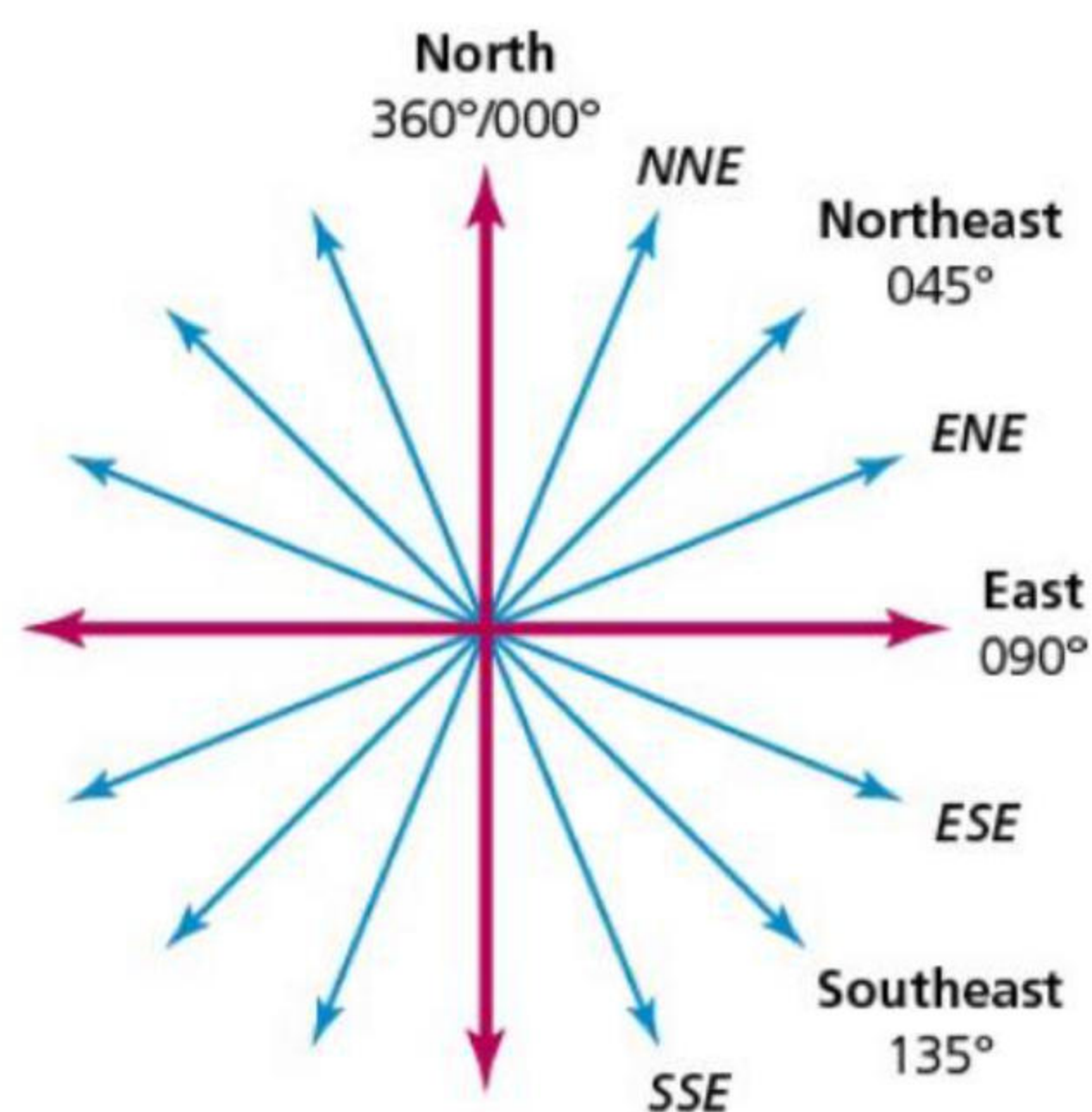
Direction

Directions can be expressed in two ways:

- » Using compass points, for example southwest. Sixteen compass points are commonly used. Some of these are shown in [Figure 11.6](#).



▲ **Figure 11.5** A section of a 1:50,000 map of Jamaica



▲ **Figure 11.6** Compass points

- » Using compass bearings or angular directions, for example 45°. Compass bearings are more accurate than compass points but they can be quite confusing.
 - Compass bearings show variations from magnetic north.
 - This is slightly different from the grid north on an Ordnance Survey map – this is the way in which the northings go.
 - True north is different again – this is the direction of the North Pole.

Gradient

The gradient of a slope is its steepness. We can get a rough idea of the gradient by looking at the contour pattern. If the contour lines are close together, the slope is steep. If they are far apart, the land is quite flat. However, these are not very accurate descriptions. To measure gradient accurately, we need two measurements:

- » The vertical difference between two points – this can be worked out using the contour lines or spot heights.
- » The horizontal distance between two places – this may or may not be a straight line (for example, a meandering stream would not be straight).

Make sure that you use the same units for both the vertical and horizontal measurements.

A spot height is a dot on the map with a number next to it which indicates the height, for example on [Figure 11.29 on page 307](#) there is a spot height at 463153 by Oxford Airport (79m).

Map symbols and map key

Symbols can be used to add a wide range of information to a map. A key should be included to allow the user to identify the symbols that have been used.

Working out a gradient

Divide the difference in horizontal distance (D) by the height (H):

$$\frac{\text{Difference in horizontal distance (D)}}{\text{Height (H)}}$$

If the answer is, for example, 10 or 5, express it as 1:10 (one in ten) or 1:5 (one in five). This means that for every 10 metres along, you rise (or drop) 1 metre; or for every 5 metres in length, the land rises (or drops) 1 metre.

Alternatively, to express the gradient as a percentage, divide the height (H) by the difference in horizontal distance (D) and multiply by 100 per cent:

$$\frac{\text{Height (H)}}{\text{Difference in horizontal distance (D)}} \times 100\%$$

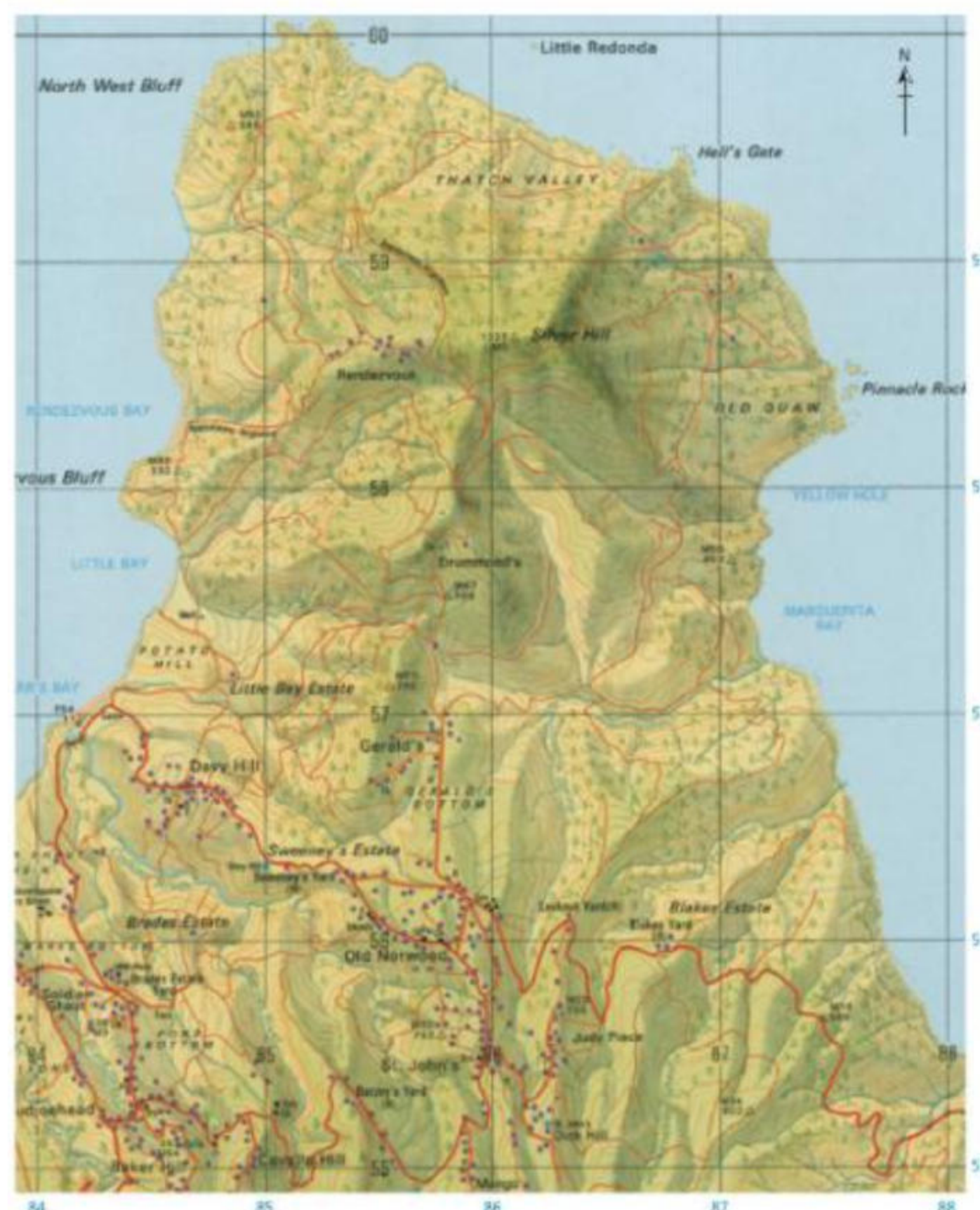


▲ **Figure 11.7** The cable car from Lago di Fedaia to Marmolada Glacier

Activities

Study the OS map shown in [Figure 11.5 \(page 294\)](#).

- 1 Calculate the distance from the school in Goodwill to the school in the middle of Dundee:
 - a in a straight line
 - b by road.
- 2 Estimate the length of the coastline shown on the map extract (to the nearest kilometre).
- 3 Estimate the length of the airstrip.
- 4 Calculate the width of:
 - a the coral in Long Bay
 - b the mangrove forest between Minto and Salt Marsh.
- 5 State the six-figure grid reference for:
 - a the two schools at Dundee
 - b Greenwood Great House.
- 6 Identify the feature found at grid reference 705023.
- 7 State the four-figure grid reference for:
 - a Chatham
 - b Davis Town.
- 8 Suggest reasons why there is an airstrip in grid square 6905.
- 9 State the direction of:
 - a Long Bay from Davis Town
 - b Goodwill from Rose Hill.
- 10 Make a copy of [Figure 11.6 \(page 294\)](#) and complete the missing compass points.



▲ **Figure 11.8** A section of a 1:25,000 map of northern Montserrat in the Caribbean

Study [Figure 11.8 \(page 296\)](#) to answer questions 11–15.

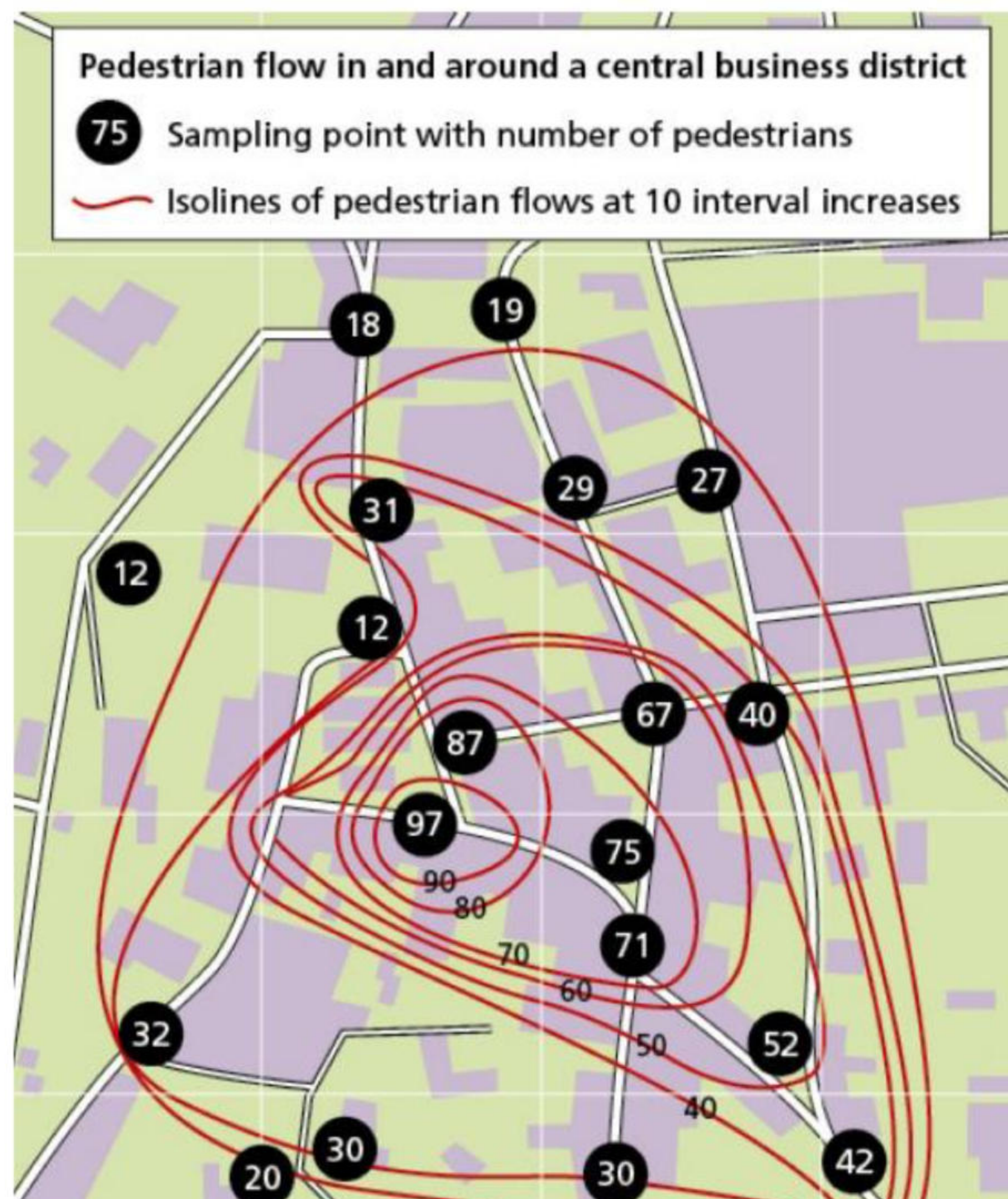
- 11 State the height of:
 - a Silver Hill (8658)
 - b Baker Hill (8455).
 (Note that the contours on this map are drawn at 50-foot intervals; assume that 3 feet equals 1 m.)
- 12 State the direction in which Little Bay (8457) faces.
- 13 Estimate how steep the slope is between Silver Hill and the coastline at Thatch Valley (8659). Measure from the peak of Silver Hill to the nearest point of the coast in Little Redonda. Express your answer as a 'one in x slope'.
- 14 Describe the relief (height and gradient) of squares 8658 (Silver Hill), 8457 (Potato Hill) and 8655 (Judy Piece).
- 15 Following an eruption of the Soufrière volcano in 1997, much of the southern third of the island was evacuated. Plans were made to develop the northern part of Montserrat. Study the map in [Figure 11.8](#).
 - a Suggest the problems of trying to develop the northern part of the island.
 - b Suggest, and justify, the best location to develop housing, services and economic activity.

Sketch maps and annotated photographs

You can label a photograph or diagram to make it very informative. It is important that you clearly label all the important features.

Many photographs that you will encounter in your studies are aerial views, which show industrial, residential, recreational and commercial land uses. In your projects, however, you are likely to use much simpler photos. If you study these carefully, you can find a number of interesting features.

Isoline diagrams



▲ **Figure 11.9** An example of an isoline map

Isoline diagrams join points of equal value on a map. They are similar to contours on an Ordnance Survey map. Isolines can only be drawn when the values under consideration change in a fairly gradual way over the area of the map. Data for quite a large number of locations are required in order to draw a good isoline map, as these maps are unsuitable for patchy data. [Figure 11.9](#) is an isoline map showing pedestrian flow in and around a central business district.

Choropleth maps

Choropleth maps can use variations in colour or different densities of black and white shading. The following steps should be followed in the construction of a choropleth map:

- » Look at the range of data and divide it into classes. There should be no fewer than four classes and no more than eight.
- » Allocate a colour to each class. The convention is that shading gets darker as values increase.
- » Now apply each colour to the applicable areas of the map.
- » Provide a key, scale and north point.

The choropleth map is a popular technique, frequently used in atlases, textbooks and many other types of publication. It can convey a lot of information in a straightforward and visually appealing way.

The main disadvantage of the choropleth is that it can appear to show abrupt changes at boundary lines, when in reality the change is much more gradual. It also gives the impression of uniformity within individual areas on the map, when in reality a reasonable degree of variation may be present. Careful selection of class sizes can reduce this problem.

Distribution maps

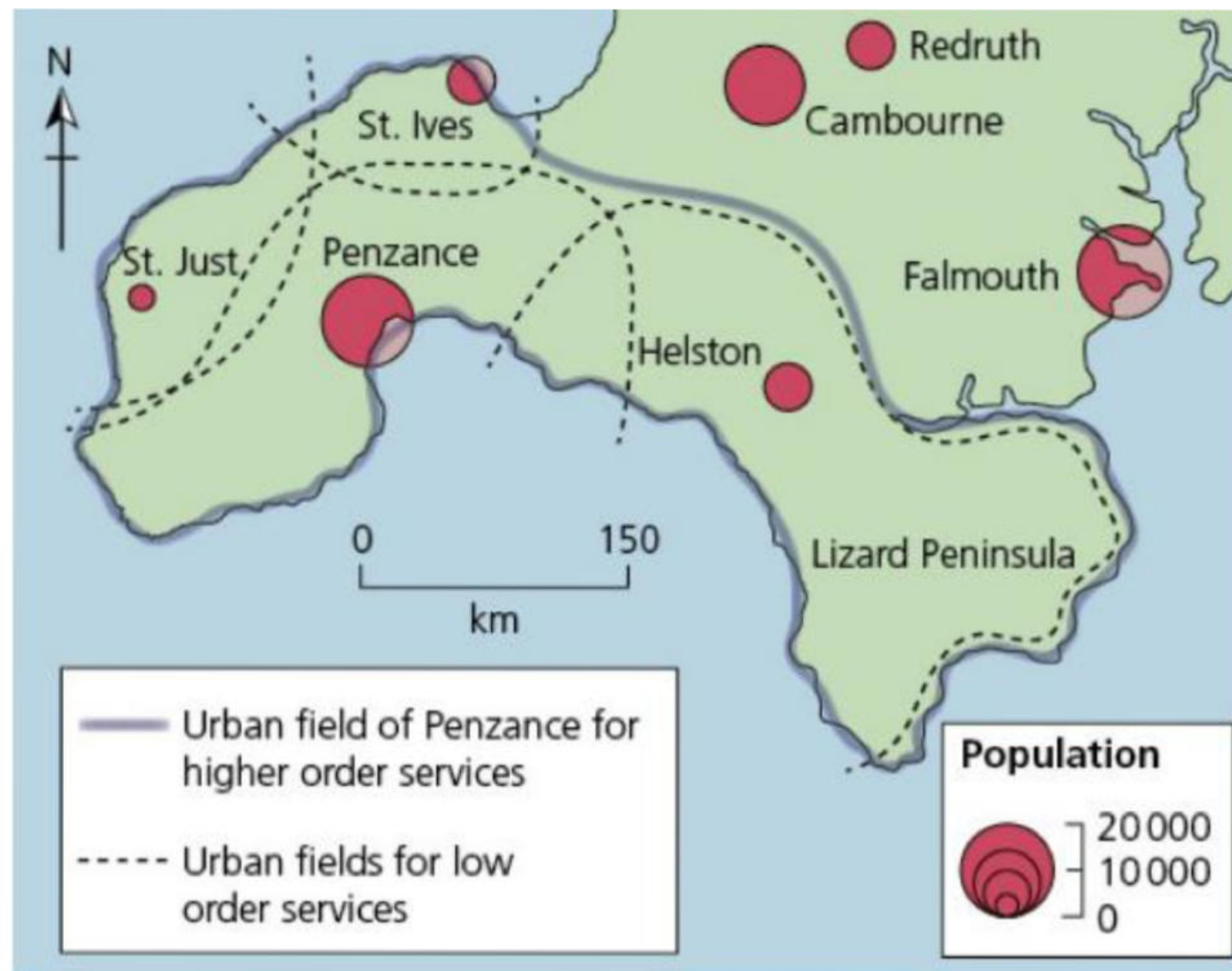
[Figure 3.10 on page 76](#) shows an example of a distribution map on the distribution of tropical rainforests.

Route maps

See [Figure 11.29 on page 307](#) for an example of a route map.

Sphere of influence maps

[Figure 11.10](#) is a sphere of influence map showing the distances people will travel for different services.



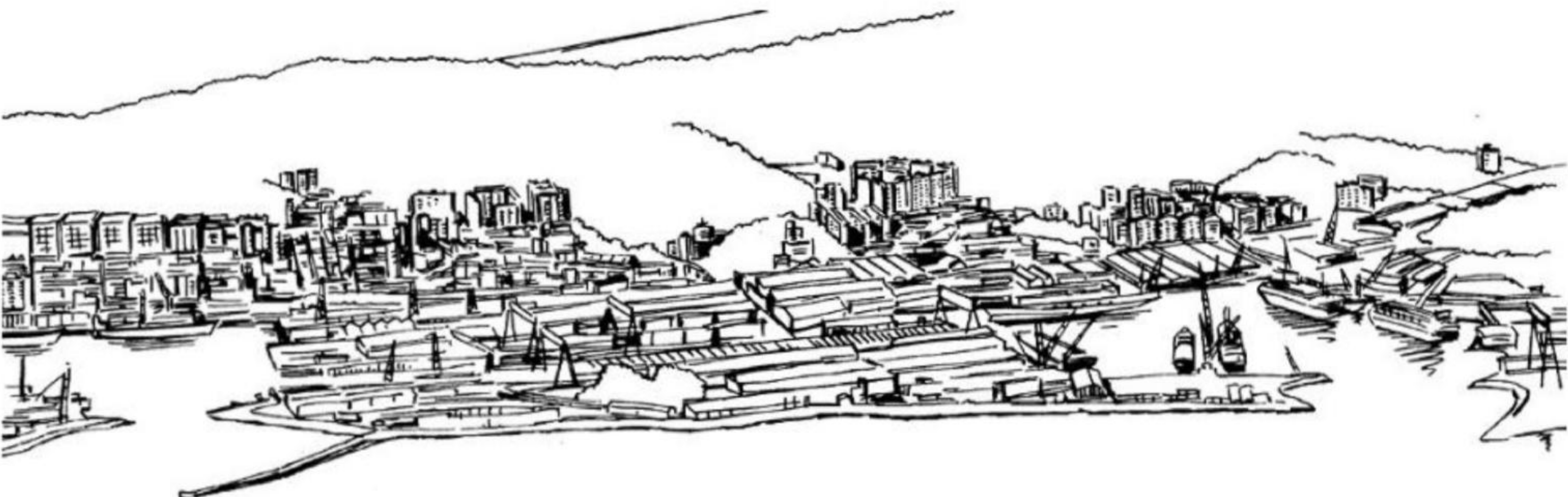
▲ **Figure 11.10** An example of a sphere of influence map

Activities

1 Study [Figure 11.11](#).



▲ **Figure 11.11** An aerial view of the Hyundai shipyard in Busan, South Korea



▲ **Figure 11.12** A sketch drawing of [Figure 11.11](#)

- a Make a copy of [Figure 11.12](#) and add the following labels in the most appropriate locations:
- Harbour wall to reduce wave energy
 - Flat land for large-scale industrial development
 - Lack of development on steep ground
 - Deep estuary allows development of port industries
 - High-rise residential accommodation
 - Large docks for ships to be repaired or built

b Add two other labels based on your own observations.

2 Study [Figure 11.13](#).

Make a sketch diagram of the resort and add the following labels:

- Purpose-built holiday resort
- Easy access to the beach
- Boat moorings
- Fine, white sandy beach
- Bay
- Lagoon



▲ **Figure 11.13** Jolly Harbour resort in Antigua

3 Look at [Figure 11.9 \(page 297\)](#), which shows an isoline map. Suggest two more examples for which isoline maps could be drawn.

Graphical skills

You may come across the techniques demonstrated below in all parts of your Geography course. Guidance is given here on how to construct graphs for coursework and the Geographical investigations, as well as how to interpret them.

Pictograms

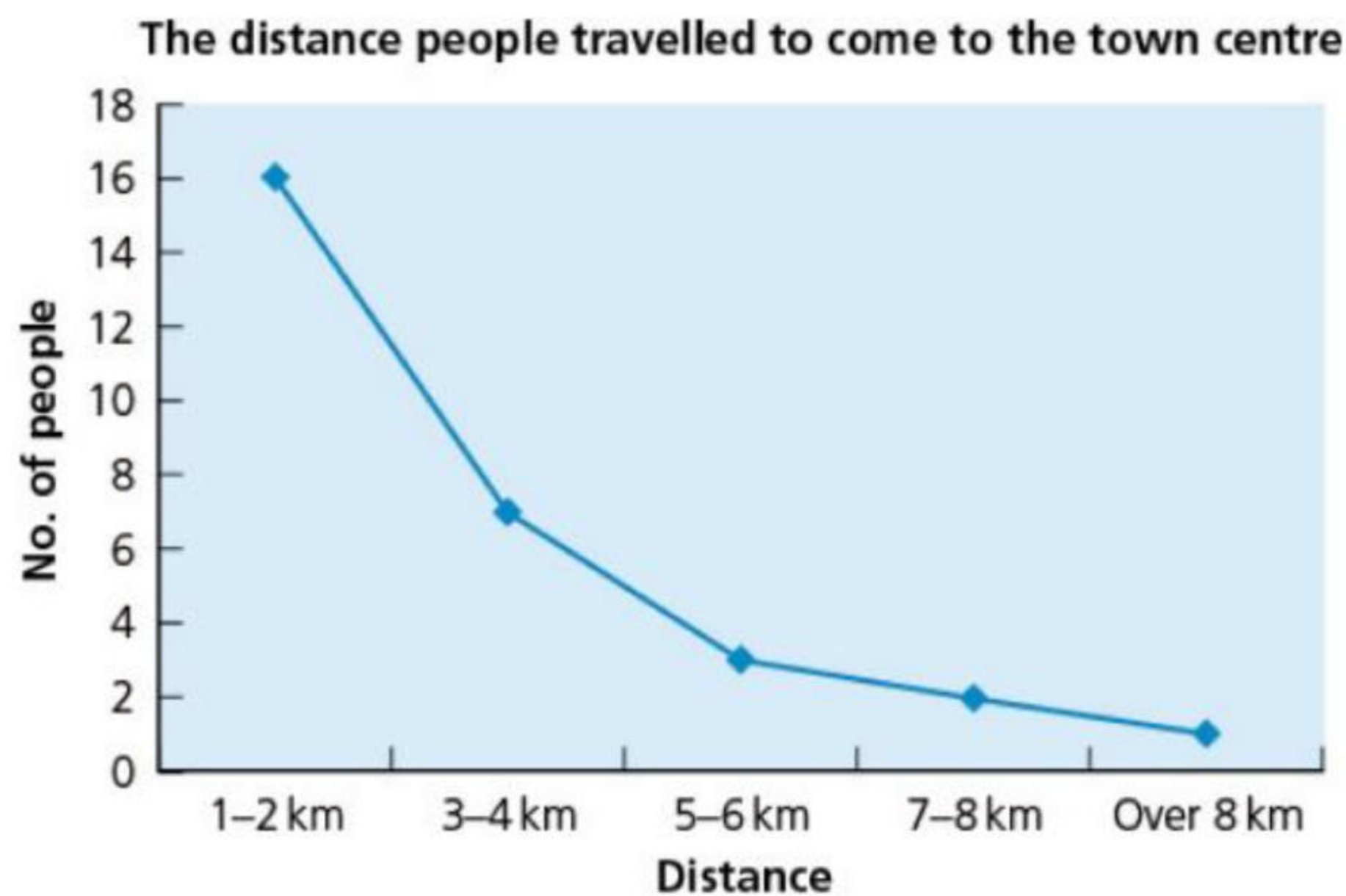
Pictograms (or picture graphs) use pictures or symbols to represent the data. The number of times a symbol occurs can represent the value or amount – in this way the pictogram acts very much like a bar chart ([Figure 11.15, page 300](#)).

Line graphs

A line graph shows points plotted on a graph, with the points connected to form a line. This type of graph is used to show continuing data. It shows the relationship between two variables. Many line graphs show changes over time, however time does not have to be one of the variables of a line graph. Examples of the use of line graphs include:

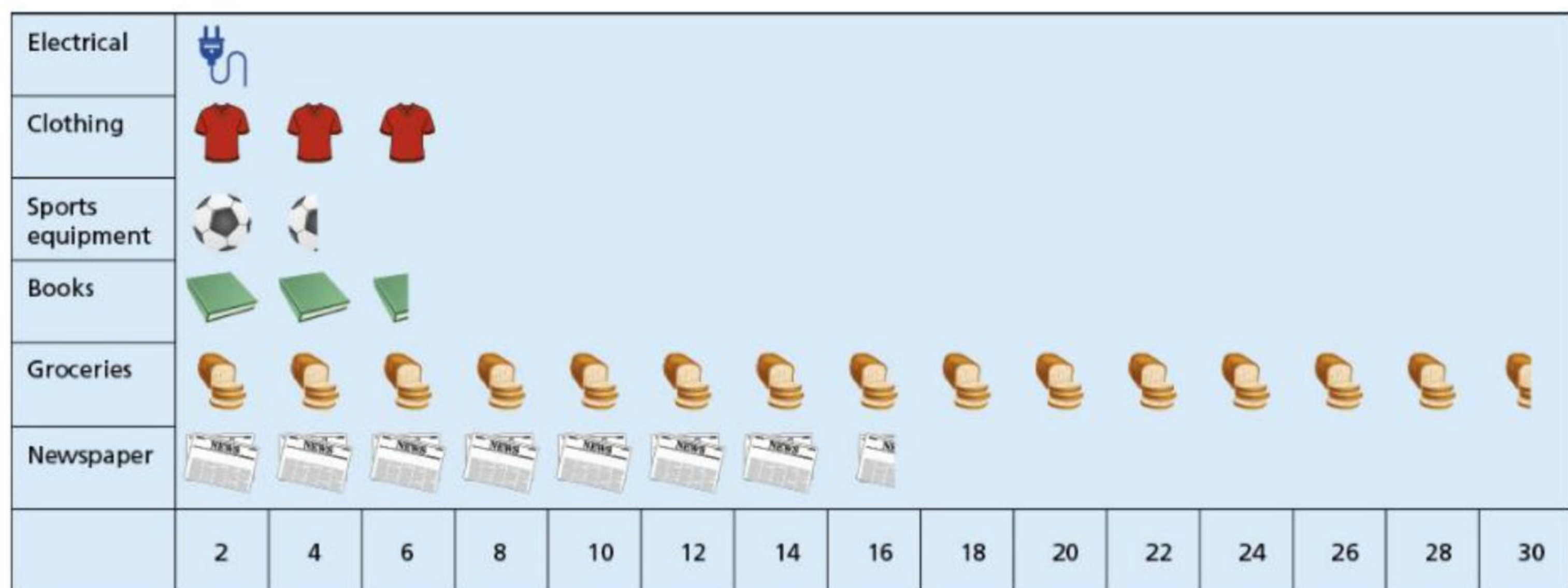
- » temperature change during the course of a day
- » pedestrian counts by time of day
- » temperature change with altitude.

The axes of a line graph should begin at zero and the variable for each axis should be clearly labelled. Be careful with the choice of scale, as this will determine the visual impression given by the graph. On [Figure 11.14](#) only one line has been drawn, but it is valid to show a number of lines so long as the course of each line is absolutely clear from start to finish.



▲ **Figure 11.14** An example of a line graph

Line graphs can be simple (showing one feature) or they may be multiple (showing many features and trends, for example the demographic transition model, [Figure 6.11 on page 132](#)). Compound or stacked line graphs show the value of each category on top of the previous ones (for example the global consumption of primary energy by source, [Figure 10.28 on page 273](#)). This is done in this way to show the value of total energy consumption, which would not be clear from a multiple line graph.



▲ **Figure 11.15** What goods did you buy today?

Activities

- Study [Figure 11.15](#). How many people bought:
 - groceries
 - clothing
 on their visit?
- Using the data in [Table 11.1](#), draw a line graph to show world population growth, 1800–2024.

▼ **Table 11.1** World population growth, 1800–2024

Year	Population (billions)
1800	1
1930	2
1960	3
1974	4
1987	5
1999	6
2011	7
2024	8

Pie charts

Pie charts are subdivided circles. These are frequently used on maps to show variations in the composition of a geographical feature. For example, in [Figure 11.16](#) they are proportional in size to the number and proportion of people employed in primary, secondary and tertiary industries. The pie chart may also be drawn proportional in size to show an extra dimension, in this case the size of GRDP.

Plotting a pie chart

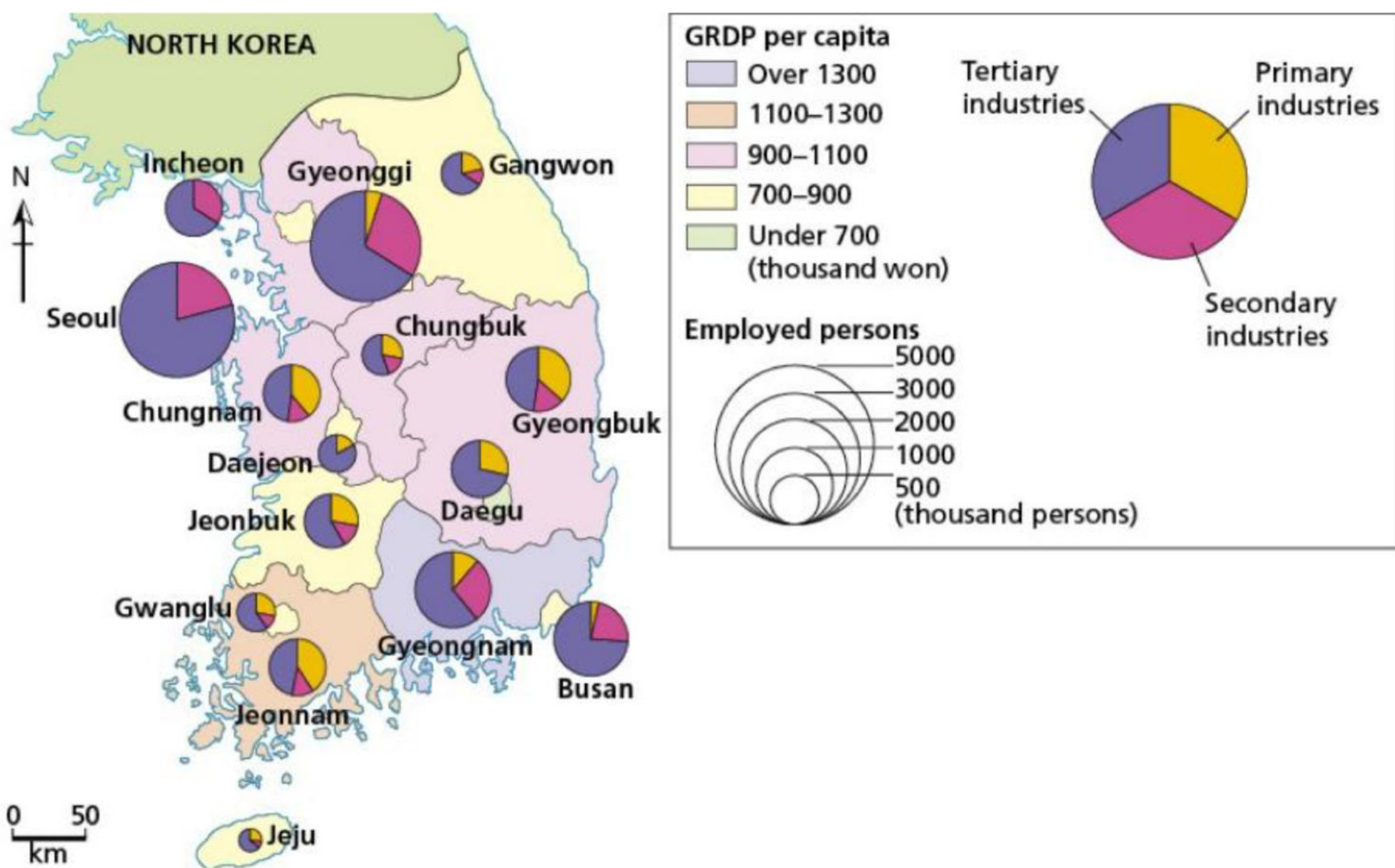
The following steps should be followed in the construction of a pie chart:

- 1 Convert the data into percentages.
- 2 Convert the percentages into degrees (by multiplying the percentage by 3.6 and rounding up or down to the nearest whole number).
- 3 Draw appropriately located circles on your base map.
- 4 Subdivide the circles into sectors using the figures obtained in step 2.
- 5 Differentiate the sectors by means of different shadings or colours.
- 6 Draw a key explaining the scheme of shadings and/or colours.
- 7 Give your diagram a title.

Bar graphs

In a bar chart, the length of the bar represents the quantity of the component being measured, for example places or time intervals. The vertical axis has a scale that measures the quantity. There are four main types of bar chart:

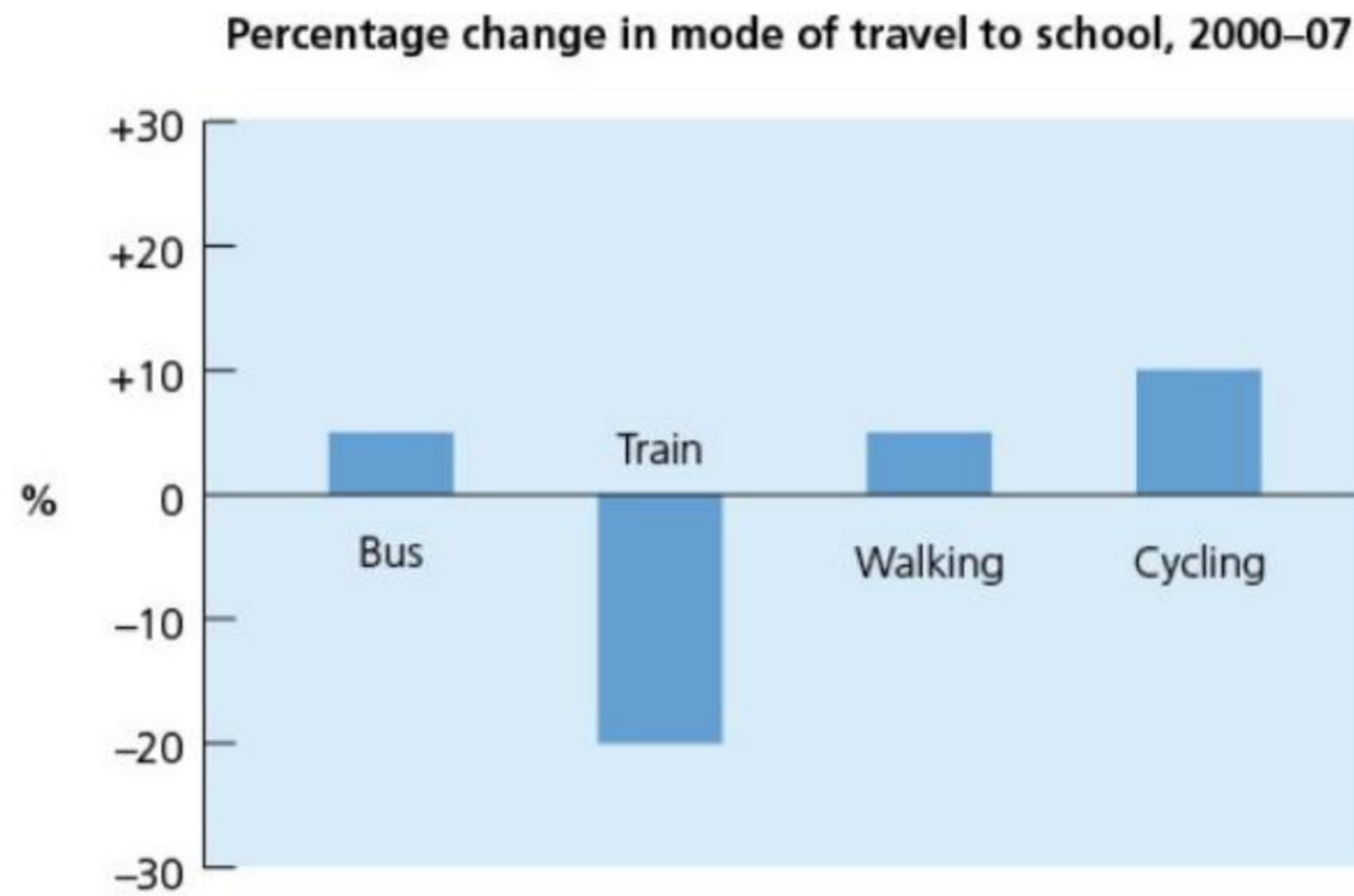
- » Simple bar chart: Each bar indicates a single factor.
- » Multiple or group bar chart: Features are grouped together on one graph to help comparison, for example [Figure 6.26 on page 148](#).
- » Compound bar chart: Various elements or factors are grouped together on one bar. The largest or most stable element or factor (the one with least variation) is placed at the bottom of the bar to avoid confusion, for example [Figure 10.28 on page 273](#).
- » Percentage compound bar chart: This is a variation on the compound bar chart. It is used to compare features by showing the percentage contribution, for example [Figure 10.30 on page 275](#). These graphs do not give a total in each category but compare relative changes in terms of percentages.



▲ **Figure 11.16** An example of proportional pie charts

Median-line bar graphs

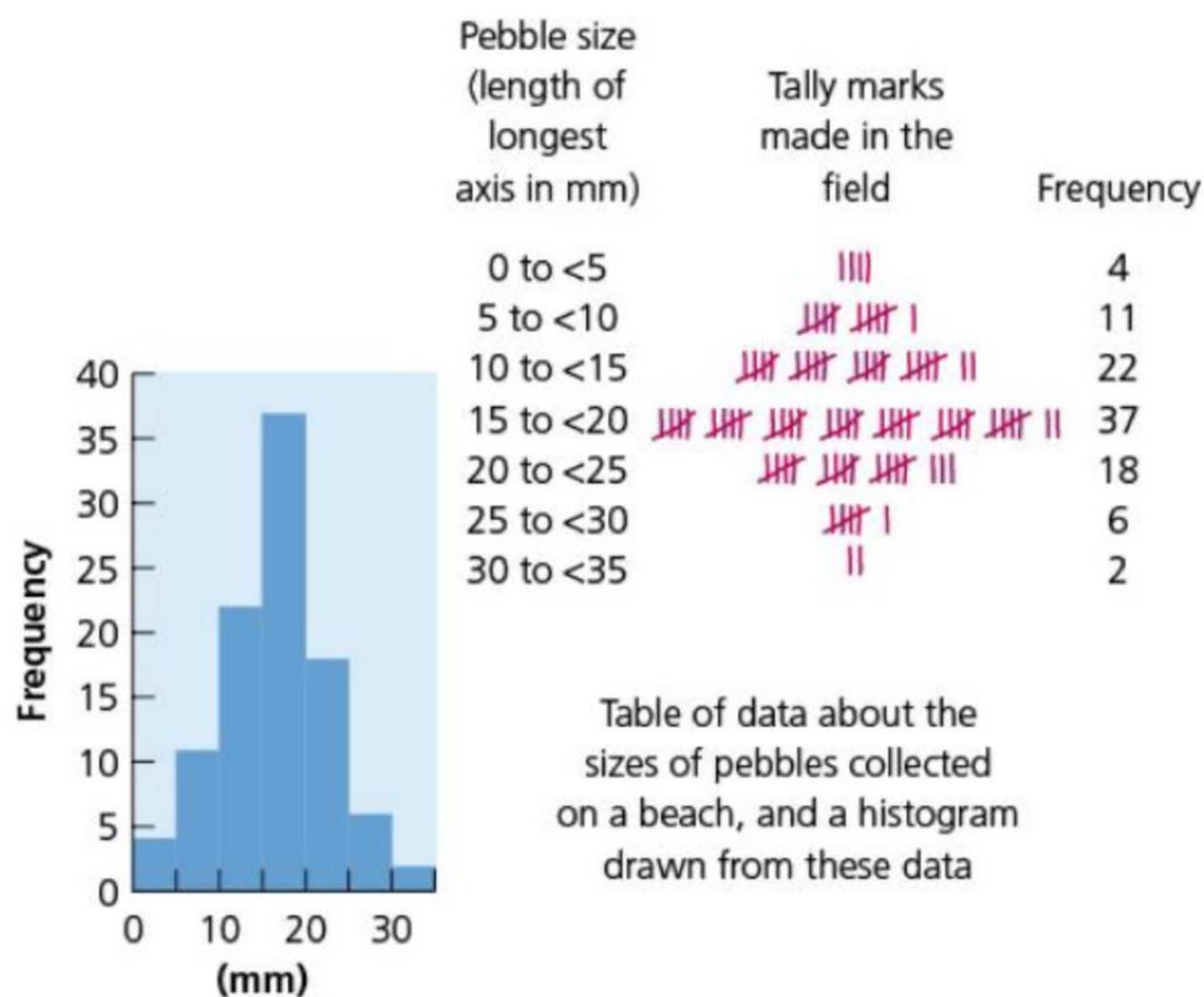
A median-line bar graph is useful when the objective is to show both positive and negative changes. The median line is set at zero, with the positive scale above the median line and the negative scale below it ([Figure 11.17](#)). This type of bar graph can create a very good visual impression. You can see instantly whether changes are positive or negative and exactly what the extent of the individual changes is.



▲ **Figure 11.17** An example of a median-line bar graph

Histograms

A histogram is a special type of bar graph ([Figure 11.18, page 302](#)) that shows the frequency distribution of data. The x-axis must be a continuous scale, with the values marked on it representing the lower and upper limits of the classes within which the data have been grouped. The y-axis shows the frequency with which the data fall into each of the classes. A vertical rectangle or bar represents each class. The bars must be continuous, without any gaps between them.



▲ **Figure 11.18** An example of a histogram and the data used

Activities

1 [Table 11.2](#) shows data for air pollution in ten cities. Construct a bar graph to show these data. Remember to add a scale and a title.

▼ **Table 11.2** Air pollution in ten cities

	City	Micrograms per m ³
1	Delhi, India	153
2	Doha, Qatar	93
3	Dhaka, Bangladesh	86
4	Kabul, Afghanistan	86

5	Cairo, Egypt	74
6	Ulaanbaatar, Mongolia	68
7	Abu Dhabi, UAE	64
8	Beijing, China	56
9	Kathmandu, Nepal	50
10	Accra, Ghana	49

2 [Table 11.3](#) shows the contribution of the main economic sectors to Korea's GDP.

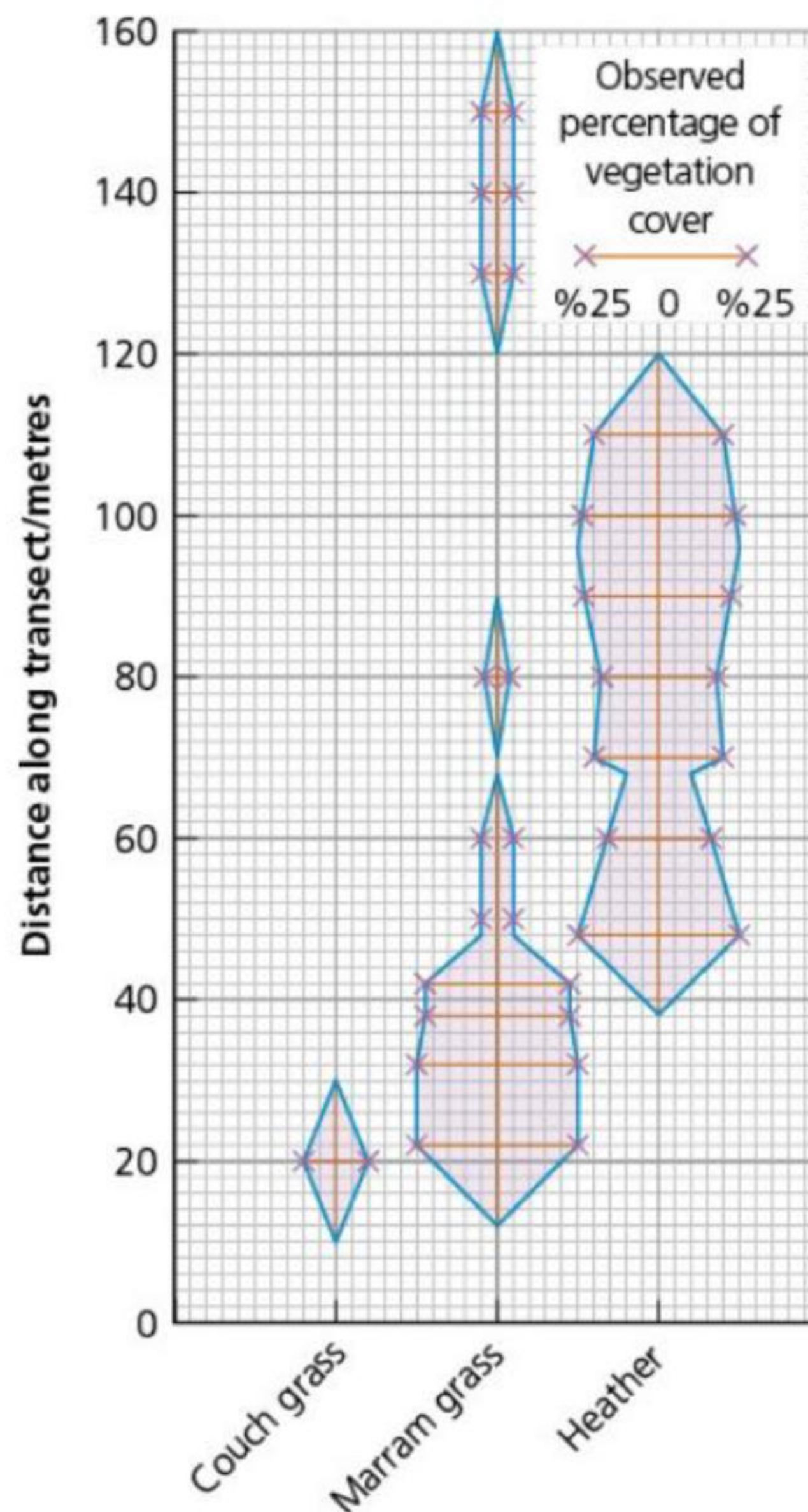
- Using the data in [Table 11.3](#), draw a pie chart showing the contribution of each economic sector to Korean GDP.
- Comment on the chart you have drawn.

▼ **Table 11.3** The contribution of the main economic sectors to Korea's GDP

Agriculture	Industry	Services
2.6%	39.2%	58.2%

Kite graphs

A kite graph is a form of line graph where the scale is split in two, so that half the values are shown on one side of the line and the other half on the other side. They are most commonly used to show vegetation distribution, for example along a sand dune or across a footpath and its surrounding area ([Figure 11.19](#)).

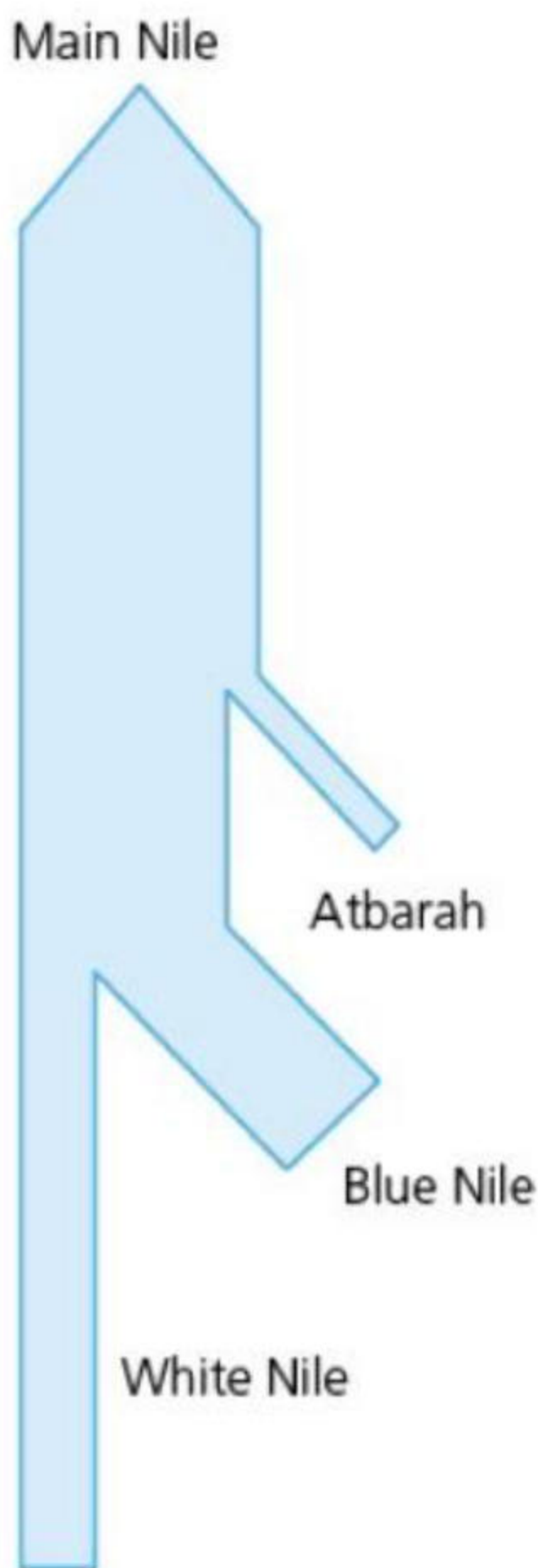


▲ **Figure 11.19** An example of a kite diagram

Flow-line diagrams and maps

Flow-line diagrams and maps are used to illustrate movements or flows ([Figure 11.20](#)). One might be used to show the variation in volumes of traffic from different smaller settlements into a larger settlement. Straight lines are used, but the width of each individual flow line is proportional to the amount of traffic it represents. Thus, a line 10 mm wide may represent 500 vehicles an hour along a road. On the same scale,

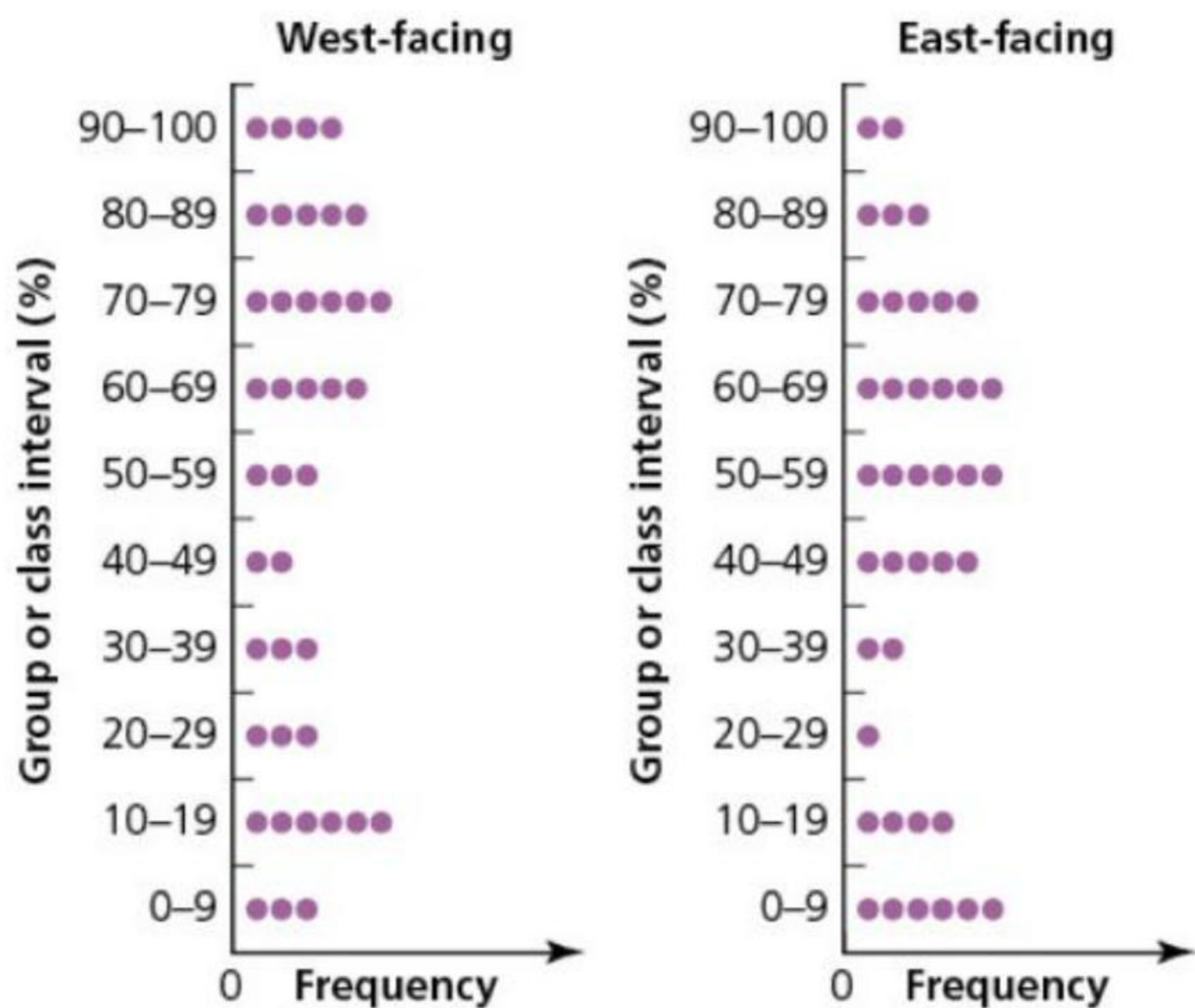
a line 2 mm wide would represent 100 vehicles an hour. Flow lines could also be used to show the number of buses coming into a town on a particular day.



▲ **Figure 11.20** A flow-line diagram showing the volume of water in the three main rivers that form the River Nile

Dispersion graphs

A dispersion graph is very useful for showing the range of a data set and its tendency to group or disperse, and for comparing two sets of data. It involves plotting the values of a single variable on a vertical axis. The horizontal axis shows the frequency. The resulting diagram shows the frequency distribution of a data set ([Figure 11.21](#)).



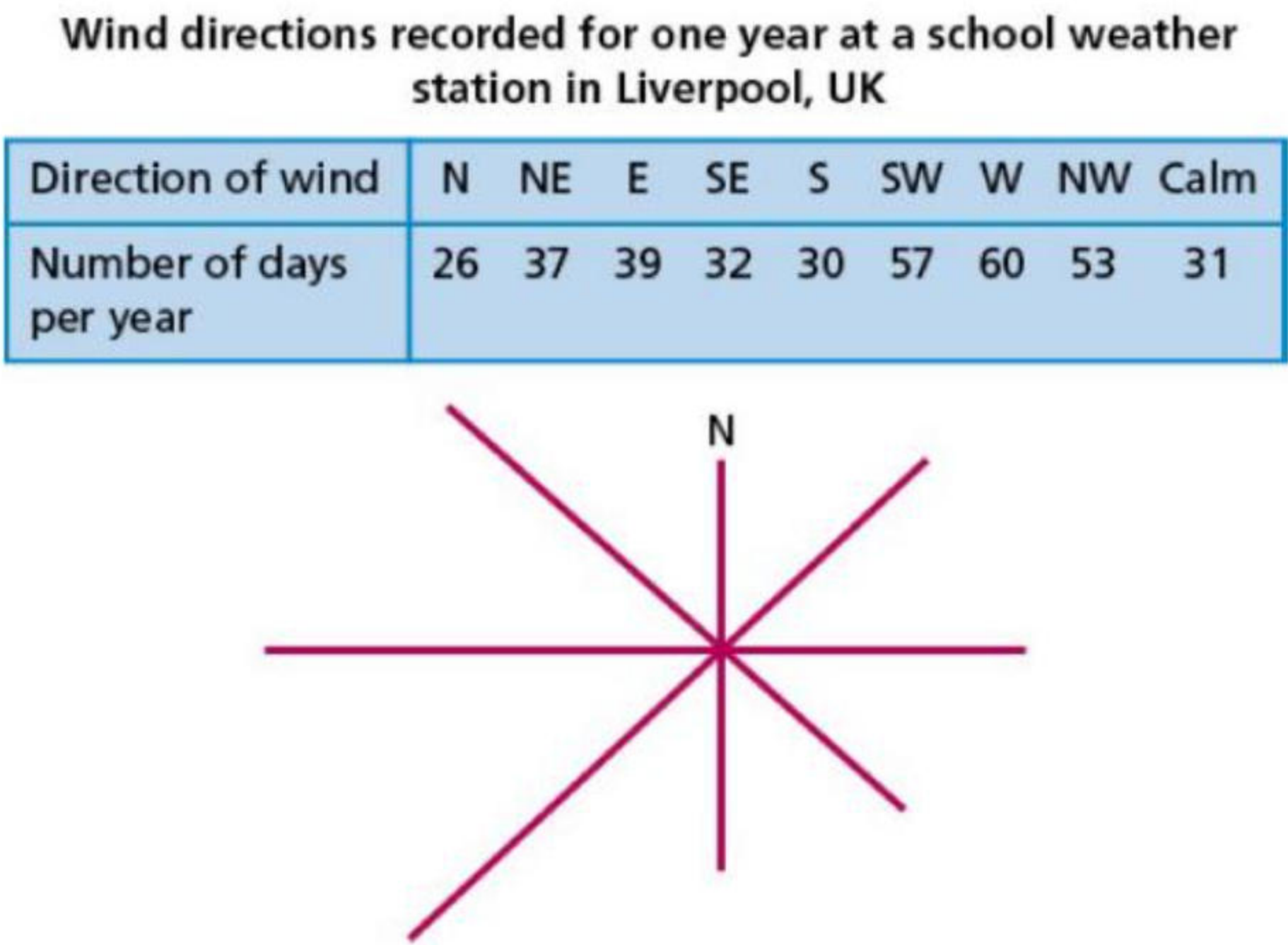
▲ **Figure 11.21** An example of a dispersion diagram showing the percentage of lichen found on west- and east-facing gravestones

Ray diagrams

There are two main types of ray diagram: **wind roses** and **desire lines**. Ray diagrams are made up of straight lines (rays), which show a connection or movement between two places.

[Figure 11.22](#) shows a wind rose diagram depicting the variations in wind direction for a certain time period. The direction of each ray relative to the centre is the direction from which the wind is blowing. Each ray is proportional in length to the number of days the wind blew from that direction.

Desire line diagrams show movement from one place to another. This type of diagram could be used to show where people live and the supermarket they use. If there are four supermarkets in an area then the rays would focus on four points rather than just one, as in a wind rose diagram. Desire line diagrams are therefore more complicated than wind rose diagrams.



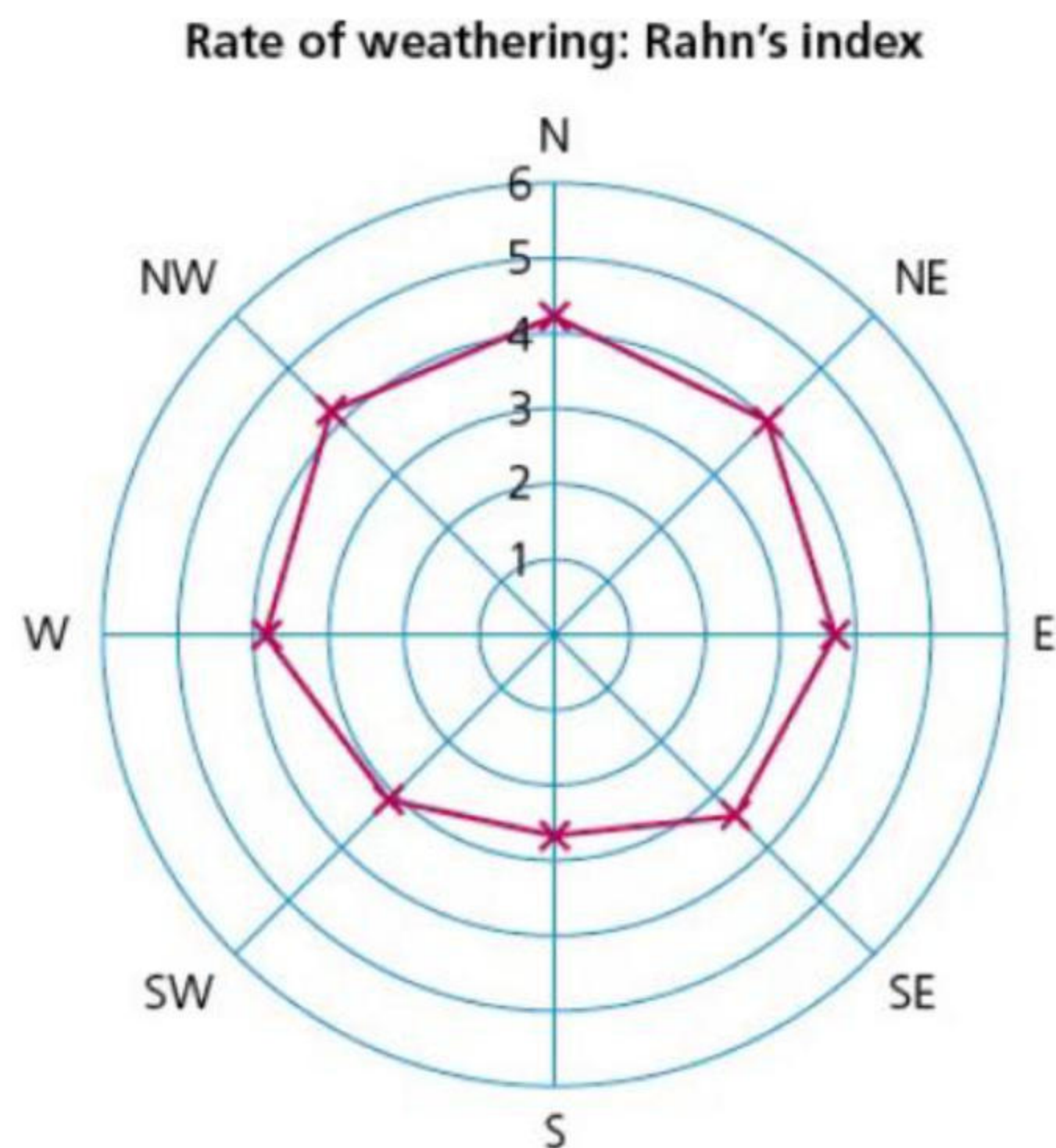
▲ **Figure 11.22** An example of a wind rose diagram

Radial (circular) graphs

Radial (or circular) graphs ([Figure 11.23, page 304](#)) can be used to plot:

- » a variable that is continuous over time, such as temperature data over the course of a year
- » data relating to direction, using the points of the compass.

The two axes of a radial graph are the circumference of the circle and the radius. Values increase from the centre of the circle outwards.



A radial graph to show the influence of aspect on gravestone weathering. The graph shows the mean of Rahn's index for each compass direction.

Rahn's index

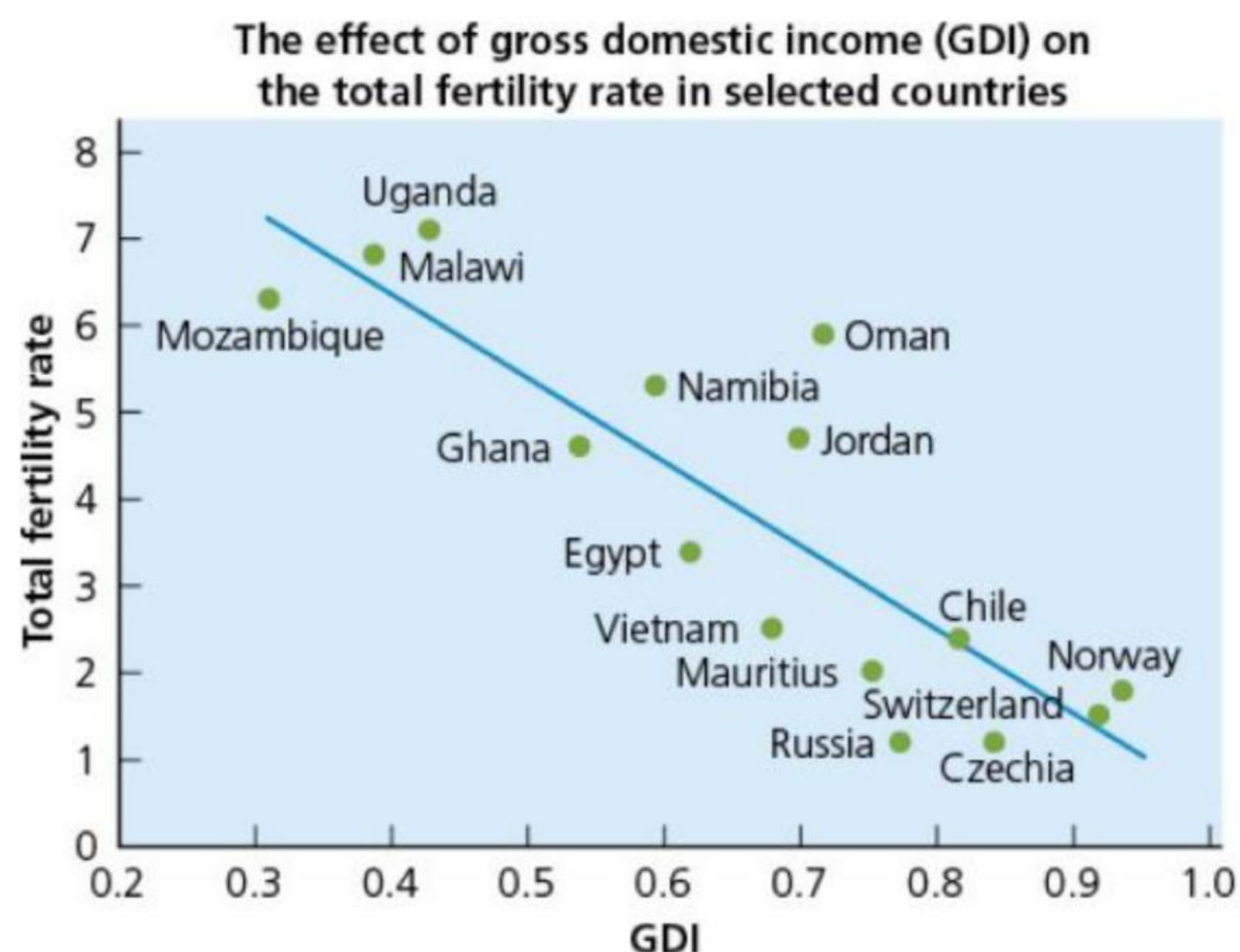
Class Description

1	Unweathered
2	Slightly weathered: faint rounding to corners of letters
3	Moderately weathered: gravestone rough, letters legible
4	Badly weathered: letters difficult to read
5	Very badly weathered: letters indistinguishable
6	Extremely weathered: no letters left, scaling

▲ **Figure 11.23** An example of a radial graph

Scatter graphs

Scatter graphs show how two sets of data are related to each other, for example population size and number of services, or distance from the source of a river and average pebble size. To plot a scatter graph, decide which variable is independent (in these examples, this would be GDI) and which is dependent (total fertility rate). The independent variable is plotted on the horizontal or x-axis and the dependent variable on the vertical or y-axis. For each data point, project a line from the corresponding x- and y-axis, and where the two lines meet, mark the point with a dot or an X ([Figure 11.24](#)).

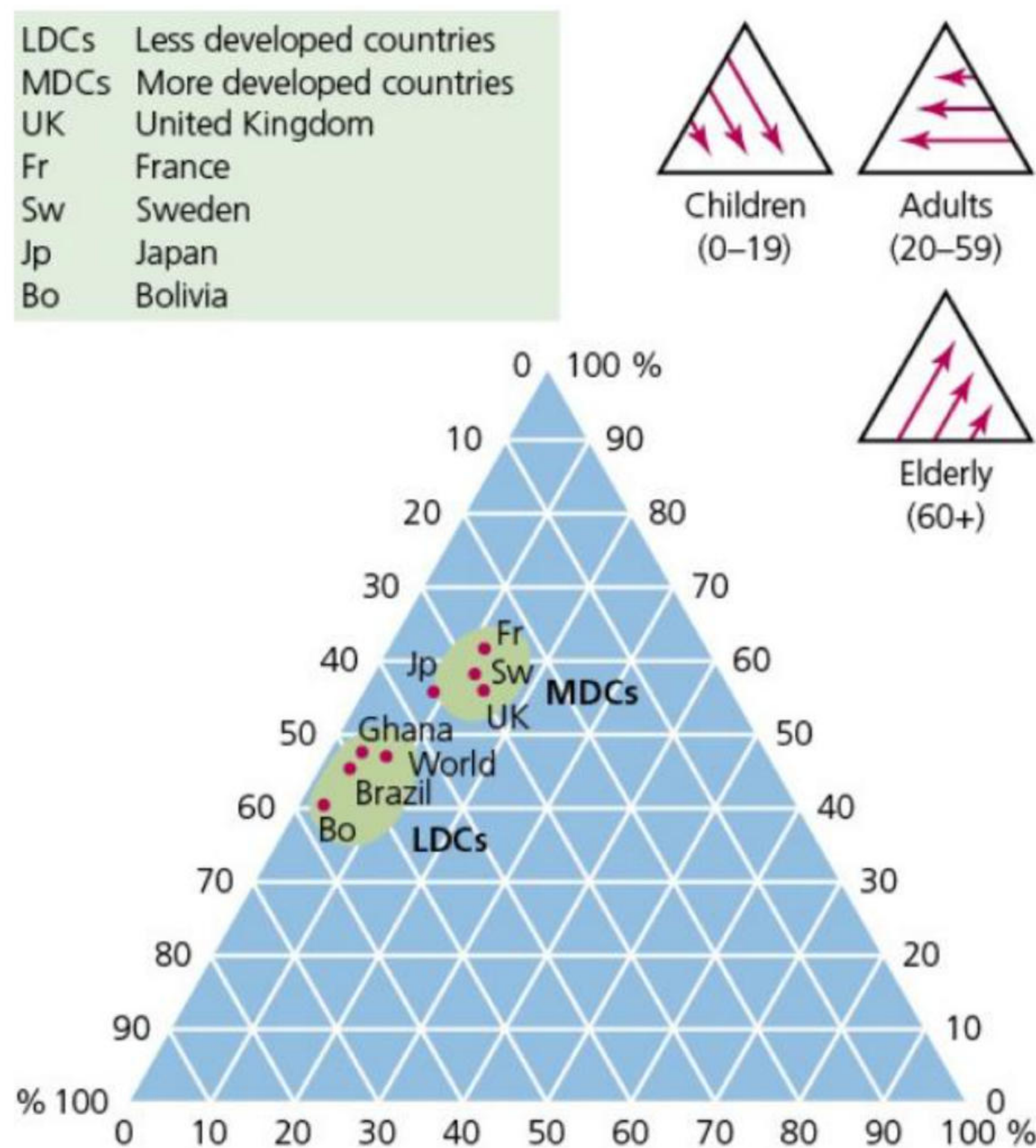


▲ **Figure 11.24** An example of a scatter graph

When all the data are plotted, a **line of best fit** is drawn. This does not have to pass through the origin. It is useful to label some of the points, for example the highest and lowest anomalies (exceptions), especially if these are referred to in any later description.

Triangular graphs

Triangular graphs are used to show data that can be divided into three parts, for example soil (sand, silt and clay), employment (primary, secondary and tertiary), or population (young, adult and elderly) ([Figure 11.25](#)).



▲ **Figure 11.25** An example of a triangular graph

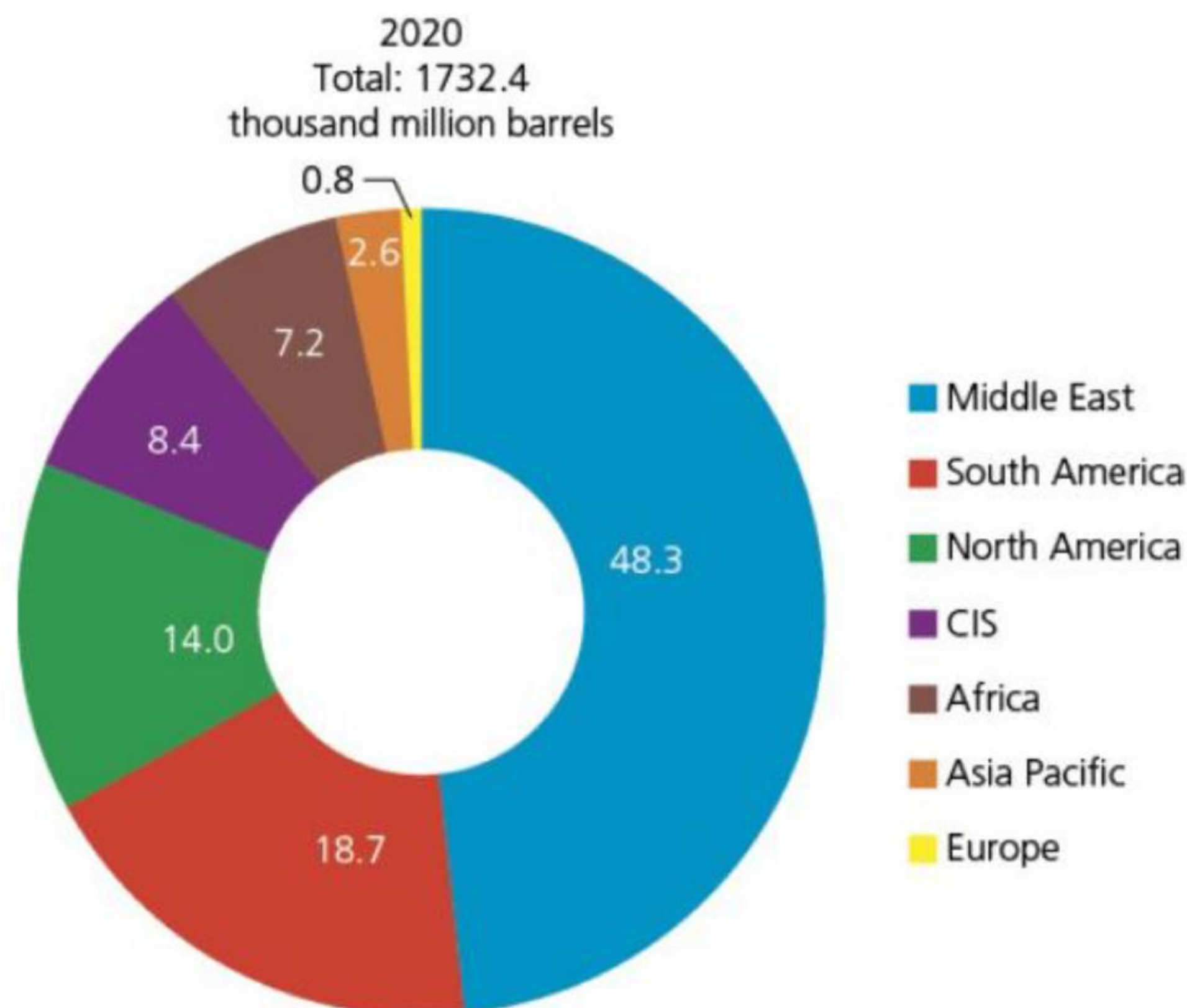
They require that the data are in the form of a percentage and that the percentages total 100 per cent. The main advantages of triangular graphs are:

- » they allow a large number of data to be shown on one graph (think how many pie charts or bar charts would be needed to show all the data in [Figure 11.25](#))
- » groupings are easily recognisable (in the case of soils, for example, groups of soil texture can be identified)
- » dominant characteristics can be easily shown
- » classifications can be drawn up.

Triangular graphs can be tricky and it is easy to get confused, especially if care is not taken, but they do provide a fast and reliable way of classifying large amounts of data that have three components.

Doughnut graphs

Another way to present data is by using a doughnut graph. An example is shown in [Figure 11.26](#).



▲ **Figure 11.26** A doughnut graph showing the distribution of proven oil reserves by region

Proportional symbols

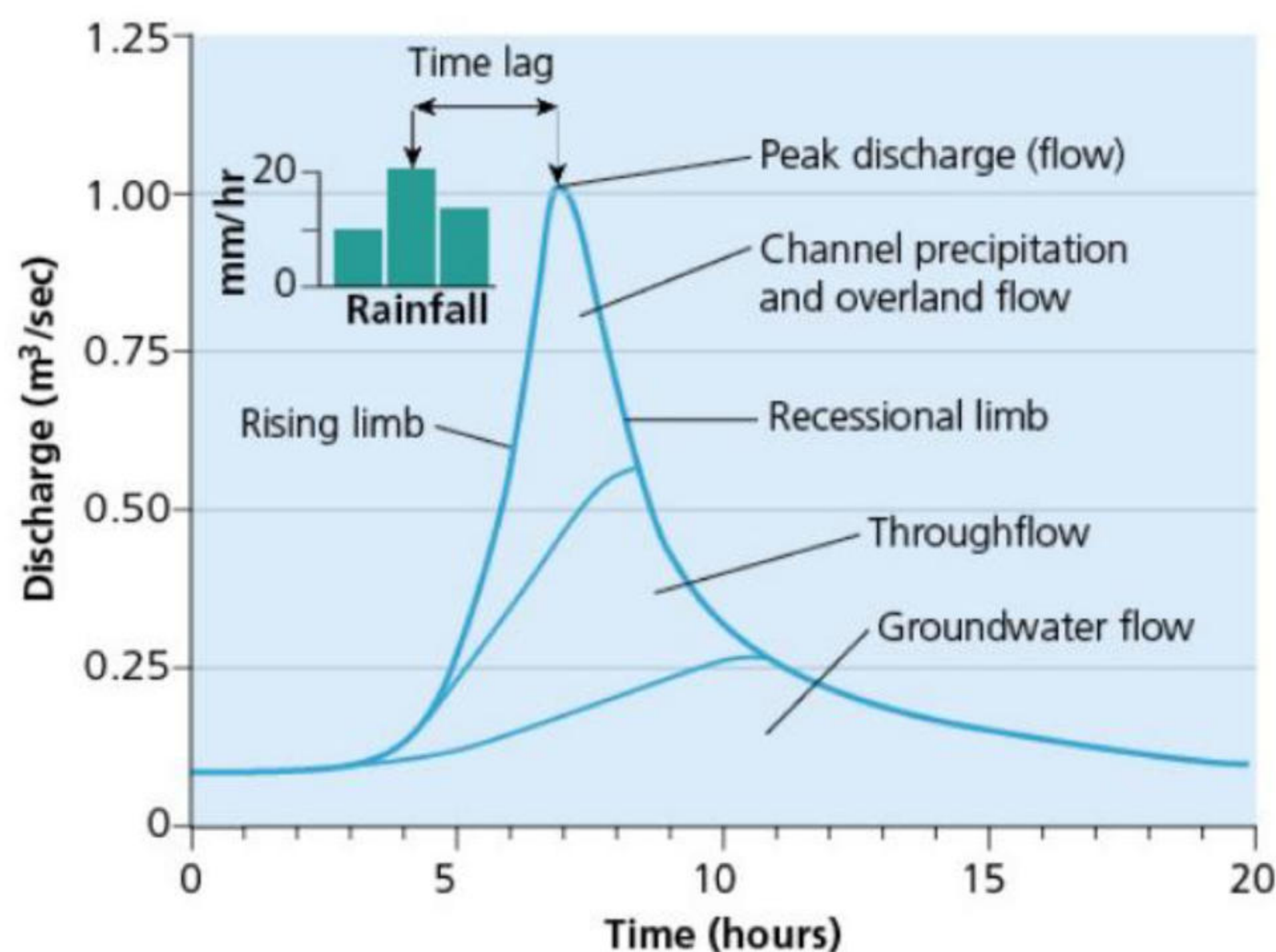
For examples of proportional symbols, see [Figure 11.16 on page 301](#).

Venn diagrams

For examples of Venn diagrams, see [Figure 1.37 on page 21](#) and [Figure 8.15 on page 191](#).

Flood hydrographs

A flood hydrograph shows how the discharge of a river varies over a short time. See [Figure 11.27](#) for an example.



▲ **Figure 11.27** A simple flood hydrograph

Activities

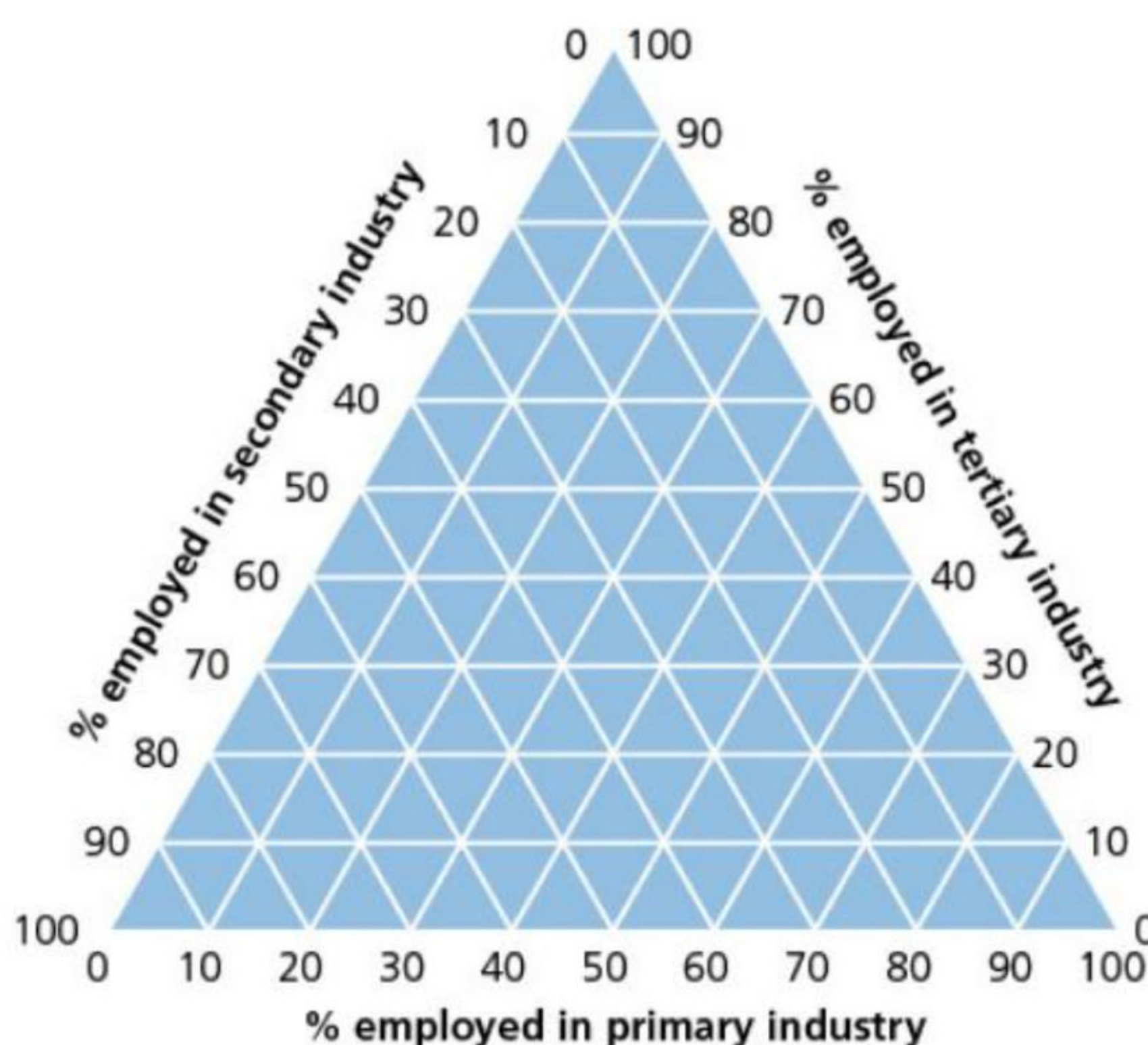
1 Construct a scatter graph using the data in [Table 11.4](#).

▼ **Table 11.4** Discharge and suspended load for ten sites

Site	Discharge (m ³ /sec)	Suspended load (g/m ³)
1	0.45	10.8

2	0.42	9.7
3	0.51	11.2
4	0.55	11.3
5	0.68	12.5
6	0.75	12.8
7	0.89	13.0
8	0.76	12.7
9	0.96	13.0
10	1.26	17.4

2 On a copy of [Figure 11.28](#) and using the data in [Table 11.5](#), show how the workforce of Korea has changed over time.



▲ **Figure 11.28** An outline for a triangular graph

▼ **Table 11.5** The percentage of the Korean workforce employed in primary, secondary and tertiary industries, 1970–2015

	Primary industries	Secondary industries	Tertiary industries
1970	50.4	14.3	35.3
1980	34.0	22.5	43.5
1990	17.9	27.6	54.5
2000	10.9	20.2	68.9
2015	5.7	24.2	70.2

- 3 Find two examples of line graphs in a Geography textbook. What do the line graphs show in these examples?
- 4 What is the difference between a histogram and an 'ordinary' bar graph?
- 5 Discuss the merits and limitations of choropleth maps.
- 6 Draw a series of proportional circles using fieldwork data or data from a textbook.

Mathematical skills

There are a number of mathematical skills that you should be able to perform. These include:

- » adding, subtracting, multiplying and dividing
- » understanding map scale and the use of scale line and representative fractions
- » understanding the terms 'mean', 'mode', 'median' and 'range'
- » using averages, decimals, fractions, percentages and ratios

- » using standard index notation (standard form), including both positive and negative indices
- » understanding significant figures and using them appropriately
- » recognising positive and negative relationships shown by scatter graphs
- » drawing and interpreting graphs from given data
- » selecting suitable scales and axes for graphs
- » using a ruler and protractor.

Map scale

If the map scale is 1:50,000, this means that every 1 cm on the map represents 50,000 cm on the ground (in reality). The representative fraction is expressed as 1/50,000 – no units are given, as it could be cm, m, km, etc.

Mean, mode, median and range

The **mean** is a type of average. It is found by totalling (Σ) the values for all observations (Σx), and then dividing by the total number of observations (n):

$$\frac{\Sigma x}{n}$$

For example, the mean number of services in eight villages was found to be:

$$\frac{(5 + 4 + 7 + 8 + 1 + 1 + 2 + 4)}{8} = \frac{32}{8} = 4$$

Activities



▲ **Figure 11.29** A 1:25,000 map of Woodstock, UK © Crown copyright and database rights 2025 Ordnance Survey (100036470). For Educational Use only

Study [Figure 11.29](#).

- 1 State the scale of the map.
- 2 Measure, in centimetres, the distance from the roundabout by Campsfield Wood (458157) to the junction of Upper Campsfield Road and Banbury Road (473173).
- 3 State the distance from the roundabout by Campsfield Wood to the junction of Upper Campsfield Road and Banbury Road:
 - a in metres
 - b in kilometres.

- 4 Estimate the length of the River Glyme from the footbridge at 437152 to Lince Bridge (443146).
5 State the distance of the longest runway at Oxford Airport.
-

There are other types of average. The **mode** refers to the group or value that occurs most often. A pattern with one peak is **unimodal**. In the example above, there are two modes: 1 and 4. Such a pattern, with two peaks (or two modes), is called **bimodal**. A pattern with three or more peaks is called multimodal.

The **median** is the middle value when all the data are placed in either ascending or descending order. In this case, we have:

8, 7, 5, 4, 4, 2, 1, 1

When there are two middle values (as in this example or whenever there is an even number of values) we take the average of the two values, which here is very easy as both middle values are 4, hence the median is 4.

The **range** is the difference between the highest and lowest values. In this example it is:

$$8 - 1 = 7$$

Decimals, fractions, percentages and ratios

A **decimal** is a number between two whole numbers. For example, the world's population in 2024 is around 8.2 billion, which means that it is more than 8 billion but less than 9 billion.

A **fraction** is part of a whole – for example, $\frac{1}{3}$ (one-third) of Borneo's rainforest has been deforested.

A **percentage** is a number or ratio expressed as a fraction of 100, using the symbol %.

- » To convert a fraction to a percentage, first convert the fraction to a decimal and then multiply by 100.
- » To convert a fraction to a decimal, divide the number above the line (the numerator) by the number below the line (the denominator). You can use a calculator for this.

Thus, in Borneo $\frac{1}{3}$, or $0.33 \times 100 = 33\%$, of the rainforest has been deforested.

A **ratio** is a method of comparing relative sizes or proportions. The deforested area in Borneo compared with the forested area is 1:2 (one-third has been deforested and there are two-thirds left).

Activity

The world's population is approximately 8.2 billion, and China's population is approximately 1.4 billion. Express China's population as a percentage of the world's population.

Standard notation

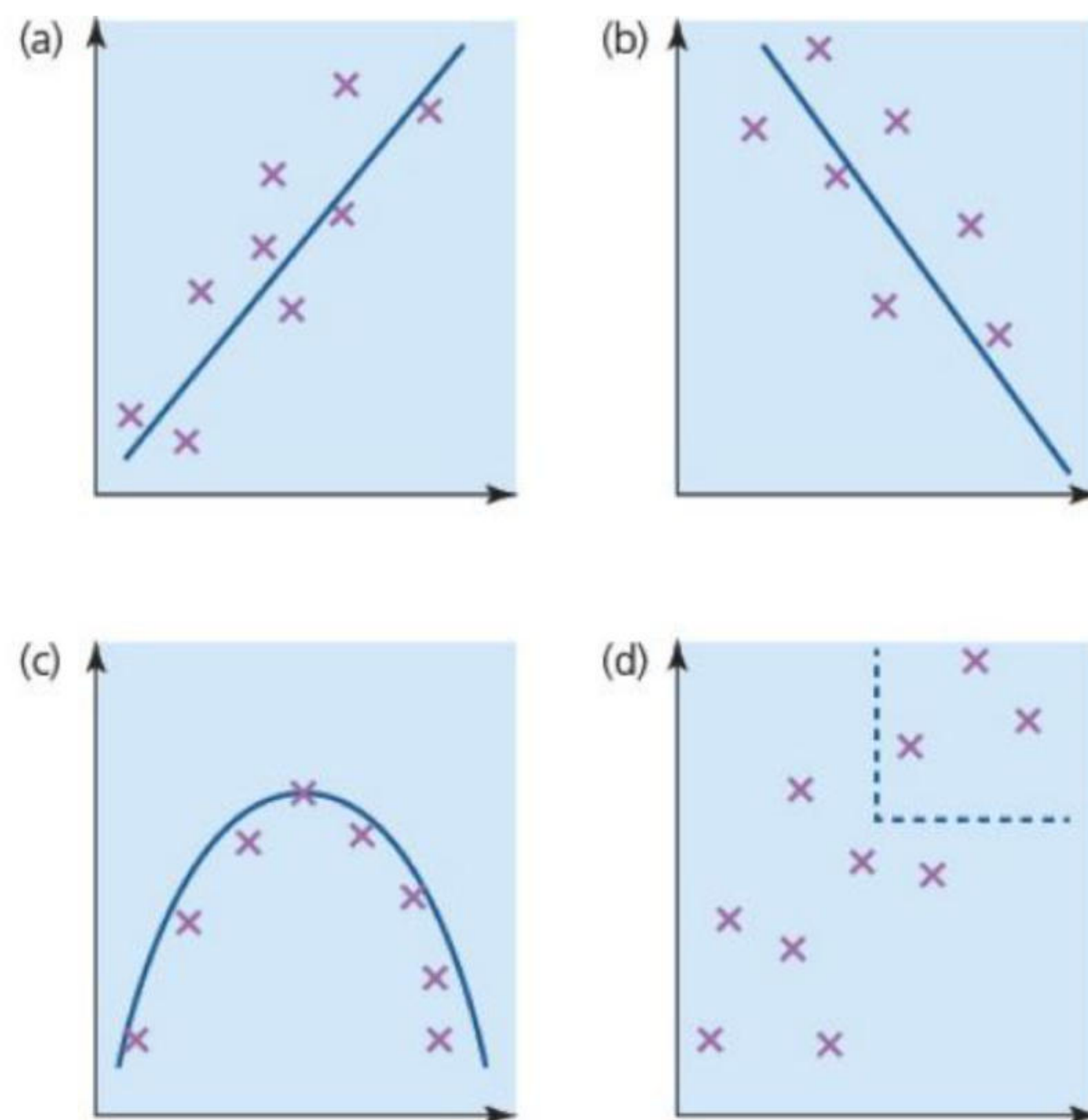
Standard notation is the number that we would normally write, for example 567. The expanded standard index notation shows that 567 is 5.67×10^2 .

A **positive index** is a power value that is positive, for example $2^2 = 2 \times 2 = 4$, or $3^3 = 3 \times 3 \times 3 = 27$.

Negative indices are powers that have a minus sign, for example $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$.

Positive and negative relationships shown by scatter graphs

Relationships between values can be investigated using scatter graphs. [Figure 11.30](#) shows a positive relationship (a), a negative relationship (b), a curved relationship (c), and a positive relationship when all values are considered but no relationship when a subset is used (d).

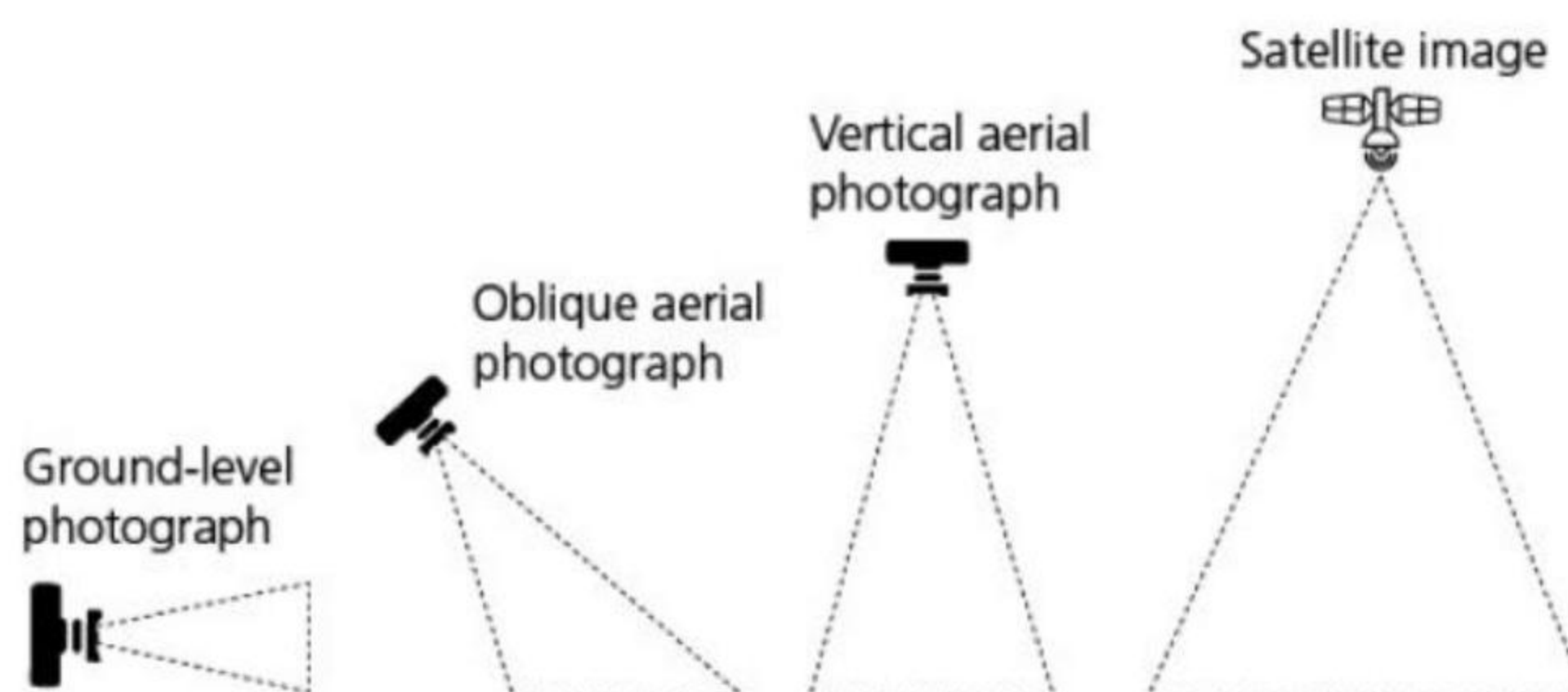


▲ **Figure 11.30** Relationships on a scatter graph

GIS and image skills

Visual images

Geographers use a wide range of visual images to record geographical features as part of fieldwork and research, and to interpret and analyse geographical issues. Photographs are an example of primary data we often collect ourselves, whereas aerial photographs and satellite images are usually secondary data. [Figure 11.31](#) and [Table 11.6](#) compare different types of image.



▲ **Figure 11.31** Photographs and satellite images

▼ **Table 11.6** Types of photographs

Ground level photographs	Oblique aerial photographs	Vertical aerial photographs	Satellite images
Used to record small-scale features; cheap and easy using digital cameras and smartphones	Used to record large-scale landscapes, using an aircraft or drone	They produce a visual image similar to a map; can show change over time, e.g. deforestation patterns	Taken from space, using visible light or infrared: tracking weather systems or monitoring planet health

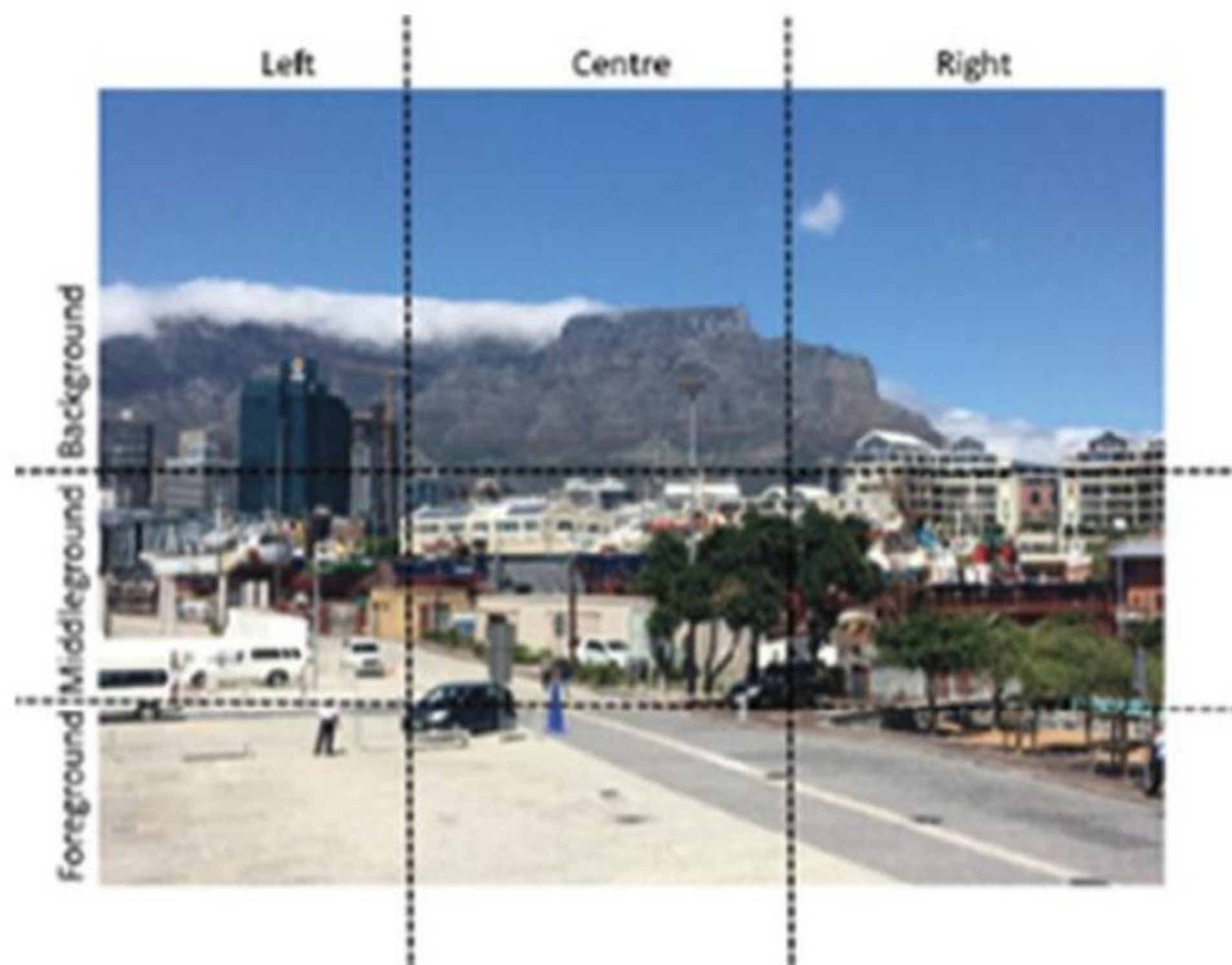
Images can be complex, which makes deconstructing and interpreting them challenging. A useful method is to visualise a grid over the image to make analysis easier ([Figure 11.32](#)). Specific parts of the image can be referred to and interpreted using the grid.

GIS

GIS stands for Geographic Information System. GIS can be thought of as a very powerful digital map that can be changed and manipulated. This contrasts with a paper map that is static, and outdated from the time it is printed. GIS computer databases contain:

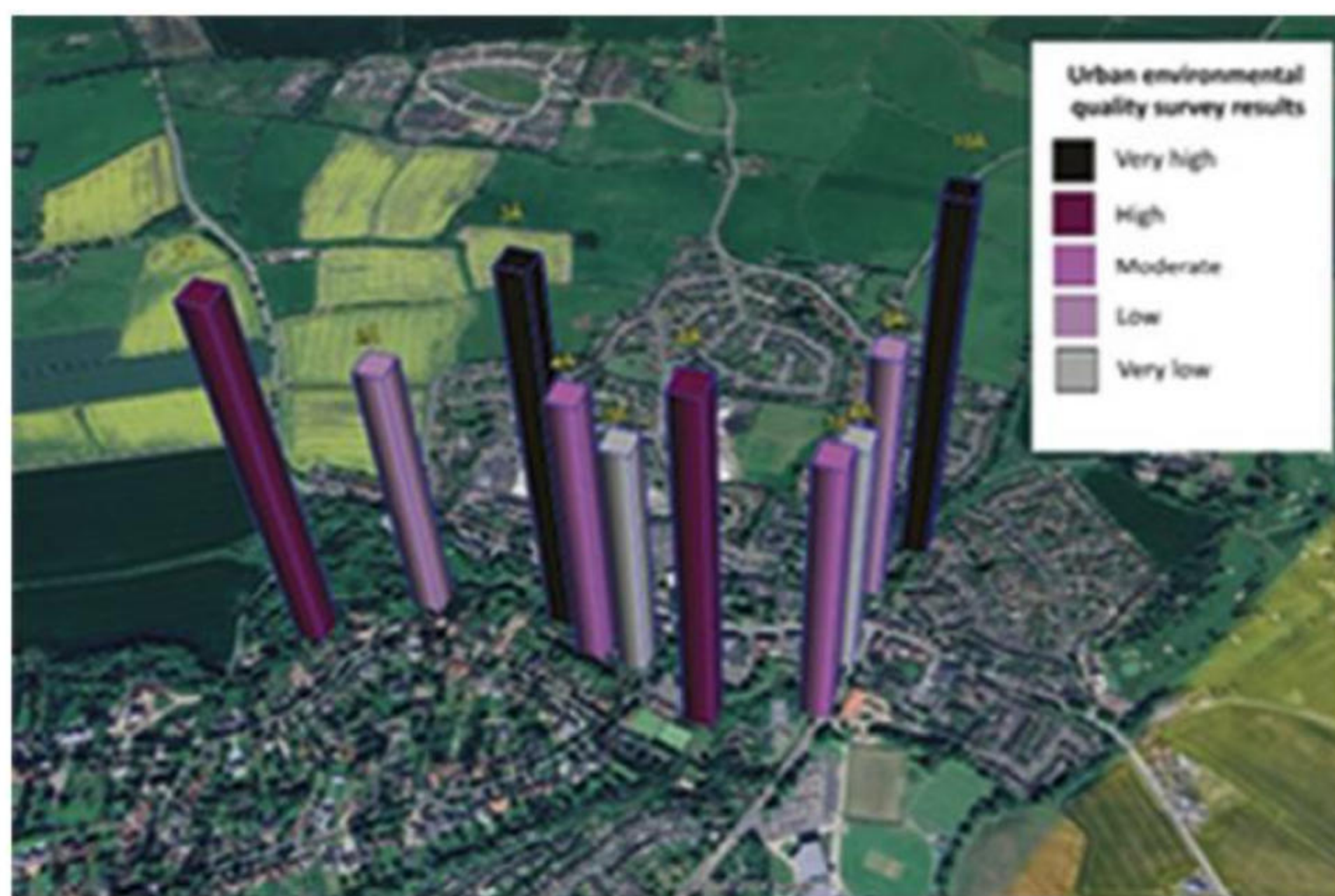
- » multiple map layers such as base maps, satellite imagery, vertical aerial photographs and digitised paper maps

» geolocated data such as photographs, data points and measurements.



▲ **Figure 11.32** An image analysis grid (Cape Town, South Africa)

Data collected during fieldwork can be added to a GIS database ([Figure 11.33](#)). GIS databases contain tools that can measure distance, area, altitude and many other features. This makes GIS a tool for analysis. Layers of information can be added and removed from a GIS to analyse patterns, trends and relationships. Both Google Maps and Google Earth are examples of GIS.



▲ **Figure 11.33** Fieldwork data results added to a GIS as coloured bars, showing the pattern of urban environmental quality in Ponteland, Northumberland

GIS has benefits compared to a paper map because it can be constantly updated and added to. It can be used to measure trends in deforestation, ocean warming and sea ice loss due to climate change. GIS is much lower cost than it was 20 years ago, and can even be used on smartphones. Limitations include the need to have some knowledge of ICT to input and analyse data, plus the scale and complexity of some GIS databases.

Cartoons

A cartoon is a diagram that may exaggerate a feature to draw attention to a particular point. What does [Figure 11.34](#) suggest about the increasing similarity of urban areas?



▲ **Figure 11.34** An example of a cartoon

Analysing text and sources

Care must be taken to think critically about sources of information used in Geography, be that text, images or data. An example is shown in [Table 11.7](#). This might seem to be an accurate data source but it is not:

- 1 The data is from 2012, so is very out of date.
- 2 The source of the data is not shown.
- 3 Only nine countries have been used, whereas Africa has 54 countries.
- 4 It is not clear why these nine countries have been chosen.

If all 54 countries are used, average income in 2012 was not \$4278 but \$2657. This is because the data chosen is biased towards wealthier countries in North Africa. There were 22 African countries with per person incomes under \$1000 in 2012, but only one is included.

▼ **Table 11.7** Average per capita income in Africa

Income per person, 2012	US\$
Angola	5083
Algeria	6096
Egypt	3059
Gabon	9348
Kenya	1289
Malawi	563
Morocco	3164
South Africa	8173
Zambia	1729
Mean income per person	4278

Fieldwork skills

Fieldwork is a very important part of Geography because it involves collecting primary data, and using secondary data sources, to investigate a geographical issue or problem. Fieldwork will help you succeed with written assessments and coursework.

Fieldwork enquiry skills, called the route to geographical enquiry, involve a structured, seven stage process to allow you to successfully complete a fieldwork investigation:

- 1 Identification of issue, question or problem – the hypothesis
- 2 Objectives of the study are defined
- 3 Collection of data
- 4 Selection and collation of data
- 5 Presentation and recording of the results
- 6 Analysis and interpretation
- 7 Making effective conclusions, evaluation and suggestions for further work.

Identification of issue, question or problem – the hypothesis

This section is the introduction to your investigation. You need to identify a topic for investigation through observation, discussion, reading or previous study. Geographical investigations begin by stating one or more hypothesis used to test the issue, question or problem. Hypotheses are the ideas you intend to test. Before you can set out your hypotheses with confidence you need to ensure that you have a good understanding of the topic (for example, sand dunes) under consideration. Studying the geographical background should ensure that you have clear knowledge and understanding of the theories or models that are used to try to explain your enquiry. You will refer back to these theories and models in your conclusion.

Examples of hypotheses:

- » Pedestrian density is highest at the centre of the CBD, and declines with increasing distance from the centre.
- » The sphere of influence of settlements increases with settlement size.
- » The pH of sand dunes decreases with distance inland.
- » Population density is higher in inner urban areas than in the suburbs.
- » Sediment size in a river decreases downstream.

For each hypothesis you investigate, you should describe what you expect to find and explain why. Within this section of the investigation you should also justify the geographical location of your inquiry. It should clearly be a good location to address the issue, question or problem you intend to investigate. Make sure that you include the area's site as well as its regional situation. Include clearly labelled location maps. Give each map or diagram used in this section a figure number. For example, a map showing the location of your study area within its wider region might be labelled 'Figure 1: The location of Studland in Dorset'. Follow this procedure for all illustrations used throughout your investigation. Also make sure that you refer to each 'Figure' in your text.

Objectives of the study are defined

Now move on to define the objectives of the study in specific terms, refining the content of the previous section. You will also make decisions concerning:

- » what data are relevant to the study
- » how the data can be collected.

It is useful in this section to briefly state the sequence of investigation you are going to follow. This should ensure that you are clear about the remaining stages of the investigation and that you tackle the route to geographical enquiry in a logical manner.

Activities

- 1 Draw a diagram to illustrate the route to geographical enquiry.
- 2 What is a hypothesis? Give two examples.
- 3 Why is it important to study the geographical background of your coursework topic?
- 4 How would you go about justifying the geographical location of your enquiry?

Risk assessment

Before you go outside and do fieldwork you must complete a risk assessment. The purpose of this is to identify risks to you and anyone else who is with you and identify ways in which the risks can be reduced. This must happen before you venture outside. Some risks are generic, such as crossing roads and being prepared for bad weather. Others are specific to an environment, such as tide times at the coast or collecting data in a river. [Table 11.8](#) is an example risk assessment.

▼ **Table 11.8** An example risk assessment (coastal fieldwork)

Risk	Cause	Consequences	Control measures	Responsibility
Slips, trips and falls	Slippery rocks and paths, uneven surfaces, algae, seaweed	Minor injuries Major injuries	Suitable footwear Avoid dangerous routes	Group leader (planning)

			/locations	Whole group
Road accidents	Busy roads, dangerous roads and junctions	Minor injuries Major injuries Fatalities	Use designated crossings and safe parking	Whole group
Adverse weather	Cold conditions and rain, strong winds	Exposure Hypothermia	Take bad-weather clothing	Whole group
Tides	Rising tides, getting wet and /or cut-off	Minor injuries Major injuries Fatalities	Check tide times Stick to timetable Carry mobile phone	Group leader (planning)
Rockfalls	Cliff collapse /landslides	Minor injuries Major injuries Fatalities	Avoid all cliffs and steep slopes	Group leader (planning)

Collection, selection and collation of data

Data can be collected on a group or individual basis, which may include:

- » fieldwork to collect primary data, such as taking measurements and undertaking questionnaires
- » gathering data from secondary sources, such as from census information and published maps, newspapers and the internet.

Make sure that you clearly explain the difference between primary data and secondary data. Use as many different techniques as possible to gather information, for example interviews, observations, surveys, questionnaires, maps and looking at figures. Describe and justify each method. Describe the use of primary fieldwork methods and in particular the method and/or equipment used to collect each type of information. Equally, describe and explain the use of secondary sources, for example parish records.

Explain clearly how you decided to use your figures, maps, answers to questions, etc. Some reasoning is necessary here – that is, justify why you used that method or source. For example, explain in detail how you questioned people, collected census figures or obtained maps. Write this up almost like the method for a scientific experiment. You can use a planning sheet here, stating when you collected data, where from, at what time, places you visited, observations you made, interviews you conducted. If you are using a questionnaire then you must justify the questions that you use, for example explain why you have recorded the age and gender of respondents in a shopping survey. Use a range of methods.

To collect data in a sound and logical way so that valid conclusions can be drawn, you should be aware of the characteristics and importance of:

- » sampling
- » pilot surveys
- » questionnaires and interviews
- » methods of observing, counting and measuring
- » health and safety, and other restrictions.

Sampling

The reasons for sampling

For many geographical investigations it is impossible to obtain ‘complete’ information. This is usually because it would just take too long in terms of both time and cost. For example, if you wanted to study the shopping habits of all 1000 households in a suburban area by using a doorstep questionnaire, it would be a huge task to visit every household.

However, it is valid to take a ‘sample’ or proportion of this total ‘population’ of 1000 households, providing you follow certain rules. The idea is that you are selecting a group that will be representative of the total population.

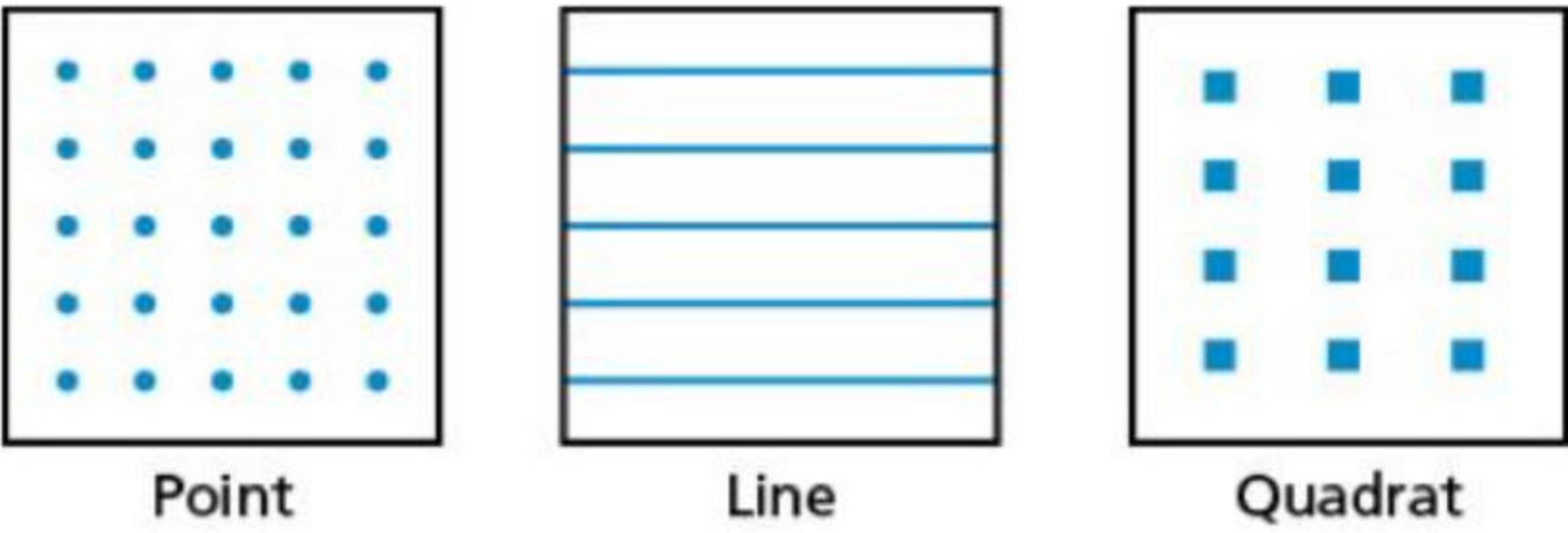
You might decide to take a 5 per cent or 10 per cent sample, which would involve talking to 50 or 100 of the 1000 households in the area. But how do you decide which 50 or 100 households to sample? There are three recognised methods of sampling that are considered scientifically valid. All three methods avoid bias, which would make results unreliable.

Sampling types

Before selecting the sampling method you need to consider how you are going to take a sample at each location. There are three ways to do this:

- » Point sampling – making an observation or measurement at an exact location, such as an individual house or at a precise six-figure grid reference.
- » Line sampling – taking measurements along a carefully chosen line or lines, such as a transect across a sand dune ecosystem.
- » Quadrat (or area) sampling – quadrats are mainly used for surveying vegetation and beach deposits. A quadrat is a gridded frame.

All three sampling types are shown in [Figure 11.35](#). Here all of the sampling types are illustrated using the systematic method of sampling. When you have read the next section you might think how these diagrams would look using random and stratified sampling.



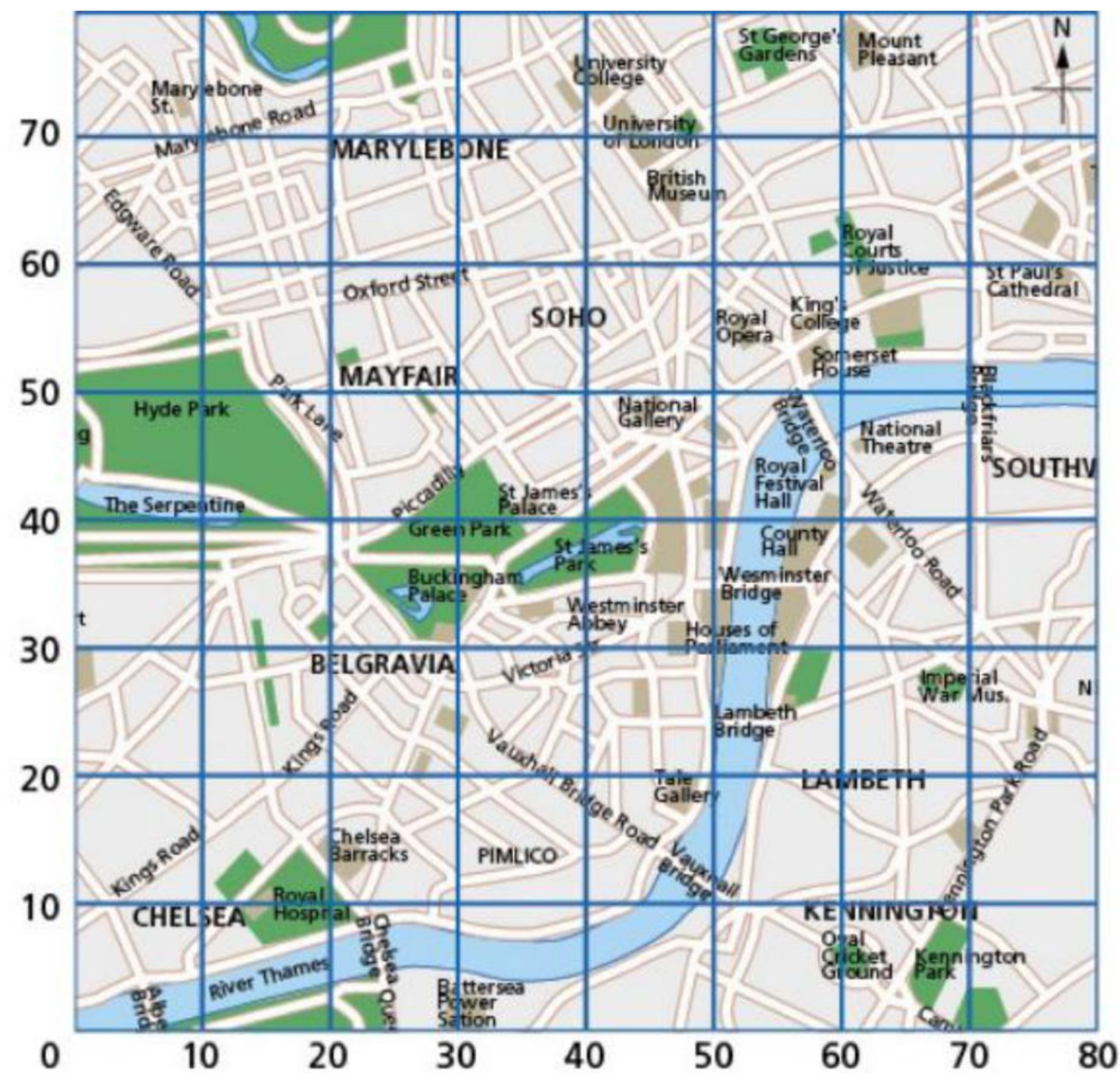
▲ **Figure 11.35** Point, line and quadrat sampling

Sampling methods

» **Random sampling** – This method involves selecting sample points by using random numbers ([Figures 11.36](#) and [11.37](#)). Tables of random numbers can be used or the numbers can be generated by most calculators. The use of random numbers guarantees that there is no human bias in the selection process.

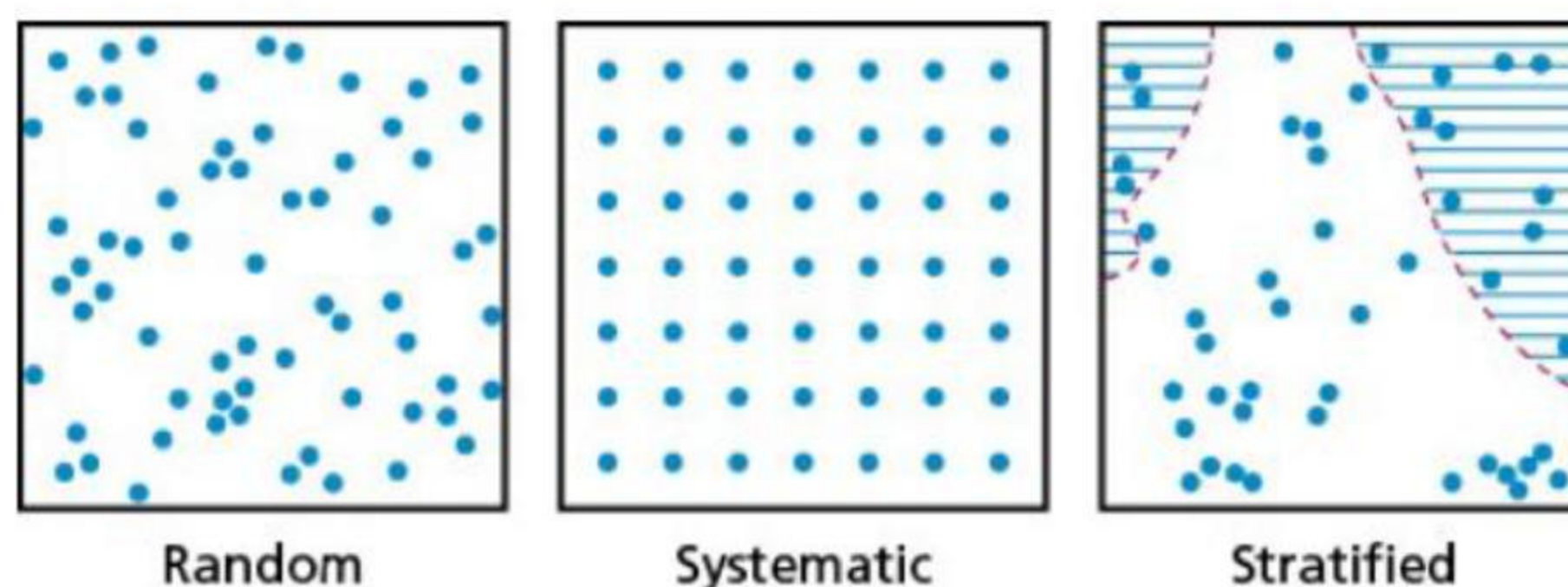
61	89	04	24	98	65	96	96
33	79	53	35	51	56	11	78
96	84	68	33	84	15	08	10
28	34	05	81	54	02	60	18
19	35	37	56	39	97	66	15
37	21	22	09	18	99	33	03
46	77	77	83	19	39	43	48
12	44	97	58	79	57	42	30
08	91	47	87	38	21	74	24
98	17	54	62	62	21	06	90
73	53	29	99	11	76	30	00
35	28	06	62	12	99	48	48
50	34	68	74	61	42	19	63
95	49	75	96	49	81	93	10
22	30	86	92	56	79	71	50
68	83	63	59	30	55	37	20
69	67	64	05	14	37	16	36
04	43	66	24	01	62	72	98
03	40	89	99	66	22	11	32
95	44	09	92	08	41	49	27

▲ **Figure 11.36** Section of a table of random numbers



▲ **Figure 11.37** How a grid and map can be used for random sampling

- » **Systematic sampling** – With this method the sample is taken in a regular way. It might, for example, involve every tenth house or person. When using an Ordnance Survey map it might mean analysing grid squares at regular intervals.
- » **Stratified sampling** – Here the area under study divides into different natural areas. For example, rock type A may make up 60 per cent of an area and rock type B the remaining 40 per cent. If you were taking soil samples for each type, you should ensure that 60 per cent of the samples were taken on rock type A and 40 per cent on rock type B ([Figure 11.38](#)).



▲ **Figure 11.38** Random, systematic and stratified sampling

Deciding on the size of the sample

The larger the sample the more likely you are to obtain a true reflection of the total population. However, this could happen with a very small sample by chance. But, equally, a small sample could give a very misleading picture of the total population.

A good rule to follow at Cambridge IGCSE with regard to sample size is to take as many samples as possible with regard to:

- » the time available
- » the available resources
- » the number of samples required for a particular statistical technique, such as Spearman's rank correlation coefficient
- » your capacity to handle the data collected (there are many computer programs available to help with this).

Pilot surveys

A pilot study or trial run can play an important role in any geographical investigation. A pilot study involves spending a small amount of time testing your methods of data collection. For example:

- » If you are using equipment, does all the equipment work and can everyone in the group use it correctly?
- » If your data collection involves a questionnaire, can the people responding understand all the questions clearly?

Activities

- 1 Why is sampling so important in geographical investigation?
- 2 Work in groups to provide outlines of different geographical investigations that would involve:
 - a random sampling
 - b systematic sampling
 - c stratified sampling.
- 3 Why might it be beneficial to conduct a pilot study prior to beginning a geographical investigation?

» If a method of sampling is used, does everyone know how to select the sample points accurately?

A small-scale pilot study allows you to make vital adjustments to your investigation before you begin the main survey. This can save a great deal of time in the long run.

Environmental quality surveys

Environmental quality surveys are used to make judgements about urban, rural or natural environments. They break down an environment into its component parts and then each component is given a score. Often a bipolar scoring system is used, which has both positive and negative numbers. [Table 11.9](#) is an example that could be used in an urban area. It scores six different component parts of the urban environment. This can be repeated in multiple locations. The scores can be averaged, and then presented in graph form or mapped using a GIS or isoline mapping.

Questionnaires

Questionnaire surveys involve both setting questions and obtaining answers. The questions are pre-planned and set out on a specially prepared form. This method of data collection is used to obtain opinions, ideas and information from people in general or from different groups of people. The questionnaire survey is probably the most widely used method to obtain primary data in human geography. In the wider world, questionnaires are used for a variety of purposes, including market research by manufacturing and retail companies and to test public opinion prior to political elections.

▼ **Table 11.9** An example of an urban bipolar environmental quality survey

Ward name: Date/time:	Location 1:	Location 2:	Location 3:	Scoring system
Building design, maintenance and condition	+2	-1		+2 Very good +1 Good 0 Neutral -2 Poor -1 Very poor
Street furniture design, maintenance and condition	+2	0		
Pavements design, maintenance and safety	+1	-2		
Road congestion and safety	0	-2		
Litter and graffiti presence /absence	+2	0		
Vegetation quantity and variety	+1	-2		
Average score:	+1.3	-1.17		

One of the most important decisions you are going to have to make is how many questionnaires you are going to complete. The general rules to follow here are similar to those for sampling, set out in the previous section. Remember, if you have too few questionnaire results, you will not be able to draw valid conclusions. For most types of study, 25 questionnaires is probably the minimum you would need to draw reasonable conclusions. On the other hand, it is unlikely you would have time for more than 100 unless you were collecting data as part of a group.

A good questionnaire ([Figure 11.39, page 316](#)):

- » has a limited number of questions that take no more than a few minutes to answer
- » is clearly set out so that the questioner can move quickly from one question to the next – people do not like to be kept waiting; the careful use of tick boxes can help this objective
- » is carefully worded so that the respondents are clear about the meaning of each question
- » follows a logical sequence so that respondents can see ‘where the questionnaire is going’ – if a questionnaire is too complicated and long-winded people may decide to stop halfway through
- » avoids questions that are too personal
- » begins with the quickest questions to answer and leaves the longer/more difficult questions to the end
- » reminds the questioner to thank respondents for their cooperation.

The disadvantages of questionnaires are as follows:

- » Many people will not want to cooperate for a variety of reasons. Some people will simply be too busy while others may be uneasy about talking to strangers. Some people may be concerned about the possibility of identity theft.
- » Research has indicated that people do not always provide accurate answers in surveys. Some people are tempted to give the answer that they think the questioner wants to hear or the answer they think shows

them in the best light.

As with other forms of data collection, it is advisable to carry out a brief pilot survey first. It could be that words or questions you find easy to understand cause problems for some people. Amending the questionnaire in the light of the pilot survey before you begin the survey proper will make things go much more smoothly.

Delivering the questionnaire

There are really three options here:

- » Approach people in the street or in another public environment.
- » Knock on people's doors.
- » Post questionnaires to people. With this approach you could either collect the questionnaire later or enclose a stamped addressed envelope. This last method is costly and experience shows that response rates are rarely above 30 per cent. Another disadvantage is that you will be unable to ask for clarification if some responses are unclear.

If you are conducting a survey of shopping habits you may want to find out if there are significant differences between males and females or between different age groups. In this case you would use a stratified sample divided by gender and the percentage of population in each age group.

The time of day may also be important. In the example given above, very few people in some age groups may be around at a certain time of day. For example, most teenagers will be in school or college at mid-morning on a weekday.

Interviews

Interviews are more detailed interactions than questionnaires. They generally involve talking to a relatively small number of people. A study of an industrial estate might involve interviews if you were trying to find out why companies chose to locate on the estate. An interview is much more of a discussion than a questionnaire, although you should still have a pre-planned question sheet. It can be a good idea to record interviews but you should ask the interviewee's permission first.

Health and safety and other restrictions

It may be sensible to work in pairs when conducting questionnaires as some people can act in an unfriendly manner when approached in the street. Working in pairs can also speed the process up considerably, with one person asking the questions and the other noting the answers. Also, be aware that shopping malls, individual shops and other private premises may not allow you to conduct questionnaires without seeking permission beforehand.

A good questionnaire

Introduction: 'Excuse me, I am doing a school geography project. Could I ask you one or two quick questions about where you go shopping?'

1 How often do you come shopping in this town centre?

More than once a week ☐

Weekly ☐ Occasionally ☐

2 How do you travel here?

Walk ☐ Car ☐ Bus ☐ Train/Tube ☐

Other _____

3 Roughly where do you live? _____

4 Why do you come here rather than any other shopping centre?

Near to home ☐ Near to work ☐

More choice ☐ Pleasant environment ☐

Other _____

5 What sort of things do you normally buy here?

Groceries ☐ Clothes/shoes ☐

Everything ☐

Other _____

6 Do you shop anywhere else, and if so where?

7 Why do you go shopping there?

8 What do you buy there?

9 Sex: M ☐ F ☐ Age (estimate) under 20 ☐

20–30 ☐ 30–60 ☐ Over 60 ☐

'Thank you very much for you help'

A bad questionnaire

Introduction: 'Excuse me, but I wonder if I could ask you some questions?'

1 Where do you live?

2 How do you get here?

3 Do you come shopping here often?

4 Why do you come here?

5 Do you buy high- or low-order goods here?

6 Is this a good shopping centre and if so, why?

7 Where else do you go shopping?

8 Do you shop there because it is cheaper or nearer to your home?

9 How old are you? _____

'Right, that's it then.'

▲ **Figure 11.39** Two questionnaires, one good and one bad

Activities

- 1 Design a questionnaire that might be used as part of an investigation into tourism in a small resort.
- 2 Briefly outline a geographical investigation in your local area that could involve the use of interviews.

Observations, counts and measurements

Field sketches

Personal observations or perceptions may form an important element of a coursework investigation. A field sketch is a hand-drawn summary of an environment you are looking at. In both urban and rural environments field sketching is a very useful way of recording the most important aspects of a landscape and noting the relationships between elements of such landscapes. The action of stopping for a period of time to sketch the landscape in front of you will often reveal details that may not have been immediately apparent.

[Figure 11.40](#) is an example of a good field sketch. This sketch highlights the important geographical features of the landscape. Key features should be clearly labelled but make sure that your sketch is not too cluttered. This will detract from the really important details. A good field sketch is a higher-level technique.

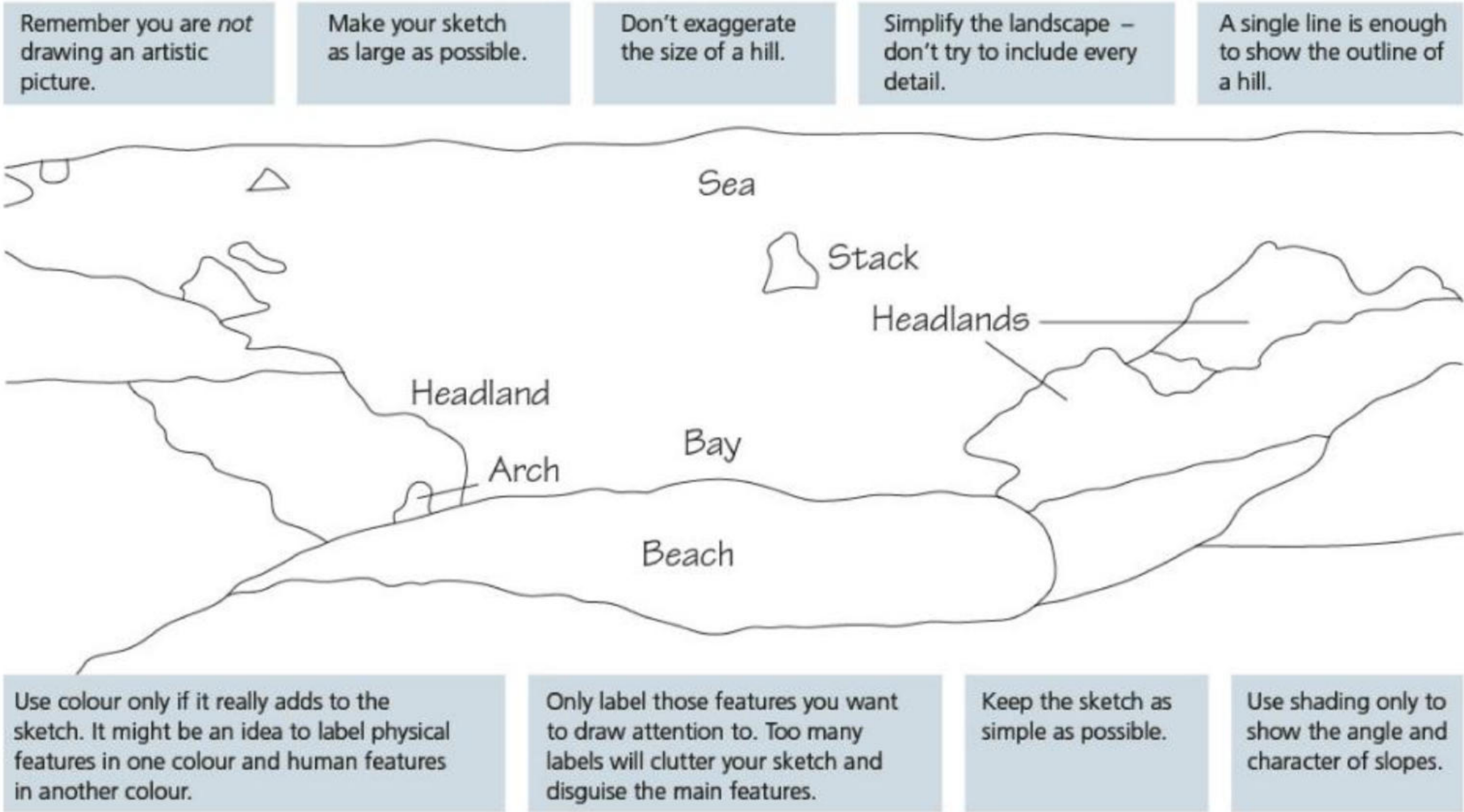
Annotated photographs

Annotated photographs should be seen as complementing field sketches rather than just being an alternative to them. Like field sketches, good, fully annotated photographs are regarded as a higher-level skill. Always record the precise location and conditions of the photographs you take. This should include a grid reference, the direction the photograph was taken in, weather conditions and time of day. Such information will make annotation quicker and easier in the long run.

An annotated photograph shows your key perceptions about a location you have visited on fieldwork. A series of such photographs might show:

- » how the type and quality of housing varies in an inner city or suburban area
- » how a river and its valley changes from source to mouth.

Annotations should be in the form of short, sharp sentences ([Figure 11.41, page 318](#)). Moderate abbreviation is fine providing the meaning of the comment remains clear. Some annotations will be just descriptive, but where the opportunity arises some explanation should also be included. Annotation can be most effective when the photograph is placed on the page in landscape format, which will allow more space for annotations on all four sides. As with field sketches, a series of annotated photographs could form a very effective part of your analysis. You should look to correlate annotated photographs with the tables and graphs showing your data analysis. Photographs are also useful to show how you carried out surveys and field measurements.



▲ **Figure 11.40** An example of a field sketch

Activities

- 1 Draw a field sketch of an urban or rural environment within easy reach of your school.
- 2 Suggest why this location has geographical interest.
- 3 Annotate a photograph of a location of interest you have visited.



▲ **Figure 11.41** A river has overflowed its banks: example of an annotated photograph

Recording tables

The most straightforward method of observation is noting whether a physical or human feature exists in an area or not. [Figure 11.42](#) is an example of a recording table showing park facilities. The objective here is to

compare the facilities in four parks before attempting to explain the differences between them. Recording is done by placing a tick in the appropriate square. Notice that there is a final column to accommodate any unexpected findings.

Pedestrian and traffic counts

It can be useful to quantify the amount of people and/or traffic in an area. This can be an indicator of how popular an area is, perhaps because it has been regenerated. It can also indicate problems such as congestion and even air pollution. When counting traffic, safety is a key consideration. Traffic and pedestrian counts can be carried out in multiple locations and the results used to construct flow diagrams. Usually, the counts are for 10 or 15 minutes. These can be multiplied by 5 or 4 to get an estimate of hourly traffic/pedestrian flow. [Table 11.10](#) is an example of a traffic count tally recording sheet.

▼ **Table 11.10** An example of a traffic count tally

Ward name:	Location /street name:	
Date /time:	Traffic moving right	Traffic moving left
Cars		
Vans		
Buses		
Lorries and other large vehicles		
Motorbikes		
Bicycles		

Park	Size (ha)	Woodland	Children's playground	Sports pitches	Tennis/basket-ball courts	Picnic site	Restaurant/ café	Boating/ fishing lake	Ornamental gardens	Pavilion/ bandstand	Toilets	Car park	Info centre/ gift shop	Other
High Lodge Forest Park	120	✓	✓			✓	✓				✓	✓	✓	Maze, jungle gym
Ditchingham Estate Park	25	✓						✓				✓		
Long Stratton Park	4	✓	✓	✓	✓					✓	✓	✓		Skate ramp
Castle Mall Gardens	2					✓	✓		✓					Viewpoint

▲ **Figure 11.42** An example of a recording table showing park facilities

Land use surveys

If your fieldwork investigation has an urban focus, particularly the CBD of a town or city, a land use survey is a useful way to measure how economic activity changes along a transect line (or lines) radiating from the CBD. Land use mapping is simple but time consuming. You will need to choose whether to only record ground-floor land use, or first-floor too (which is more time consuming). You need:

- 1 A detailed base map showing individual buildings at a scale of 1:1250 or less (such as OpenStreetMap).
- 2 A transect line running from the centre of the CBD outward.
- 3 A land use key, such as RICEPOTS (see [Figure 11.43](#)) and coloured pencils.

Some investigations focus on delimiting the CBD, in other words finding the edge of the CBD to determine its spatial extent or sphere of influence. This can be done by:

- » using a land use survey to identify where residential land use takes over from commercial and retail land use
- » using pedestrian and traffic counts to map changes in volume
- » measuring building height (number of storeys) to find where land values decrease so most buildings are 1–3 storeys, not higher
- » using a noise meter, which measures noise in decibels, to measure traffic and other noise.

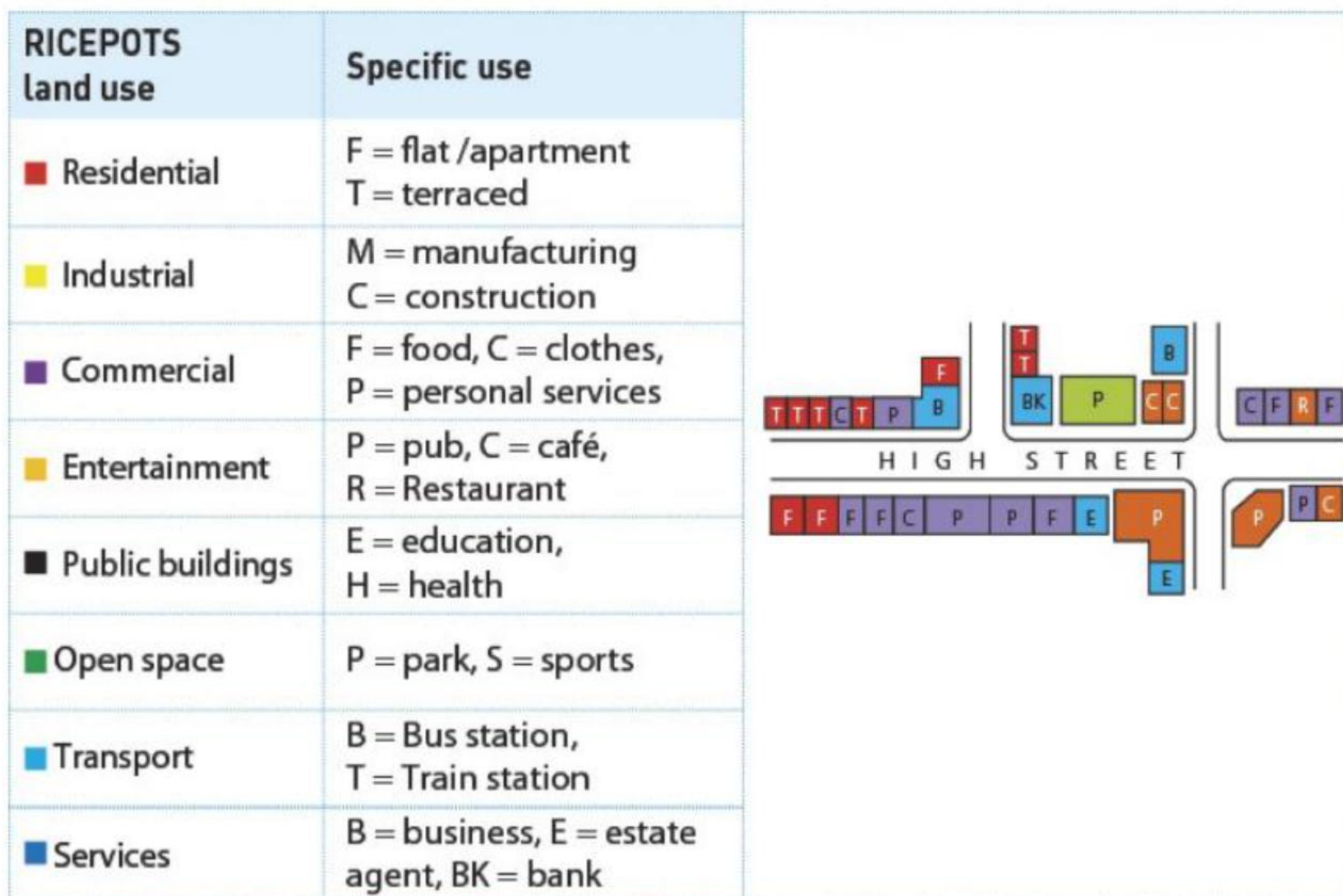
Activity

Produce a recording table that could be used as part of a geographical investigation in your local area.

Presentation and recording of results

A wide variety of graphical techniques can be used to present geographical data. The skill is in choosing the best type of graph for the particular data set under consideration. You should also consider the size of any graph or diagram you use. It is important that the labels of axes and all other information can be clearly read.

It is important to integrate all maps, graphs, photographs and diagrams with the text. The most elementary way of doing this is to use a sentence such as 'Figure X is a line graph showing temperature change in my garden'.



▲ **Figure 11.43** A RICEPOTS land use survey key and example

Analysis and interpretation

Here you will analyse and interpret your findings on the issue/question/problem set out in the first section of your investigation. This will be done with reference to relevant geographical concepts.

You need to describe the patterns in data presented in your graphs and tables of results.

- » After each graph or technique, describe fully the results or trend or association (using simple descriptive statistics). What do your results tell you? Describe your findings in detail by quoting the evidence from your methods of analysis.
- » Do the graphs and other techniques used help to answer the question set?
- » Make comments to link the data. For example, show how one diagram, graph or map relates to others.
- » Where relevant, consider the values and attitudes of people involved.

Making effective conclusions

Using the evidence from the data you should be able to make judgements on the validity of the original hypothesis or aims of the assignment. Compare the results of the data analysis against standard models and theories.

Link what you have discovered in your enquiry to what you have studied in the syllabus. For example, if you have looked at shopping, have you talked about high-order and low-order shops, shopping hierarchies and other relevant concepts? If your investigation is on leisure, have you linked this to the amount of leisure time people have, spheres of influence of leisure centres, how accessible places are and other key aspects of the topic? These things should be mentioned briefly in the first section of your assignment and need to be discussed now in relation to your findings.

Having described your results, you need to explain and discuss them. Why have you arrived at such results? Do they confirm (accept) or refute (reject) your hypotheses?

Evaluation and suggestions for further work

Your investigation is likely to be slightly less than perfect, and you need to show the examiner that you are aware of this. It is important to show an awareness of the limitations of methods, results and conclusions.

There may be limits to where and when you could carry out a survey, or the number of people you could interview. There may be constraints in terms of expense, in terms of the equipment you are able to use, or

how often you can visit the fieldwork location(s). The last item can also be due to a busy school timetable, which can mean your time at a fieldwork location is limited. Investigations in physical geography may be hampered by unexpected conditions. For example, river measurements may be affected by floods or a drought.

You need to make an assessment and state whether or not such limitations have impaired your investigation. If they have, then your results are likely to be compromised, as will any conclusions that you have drawn. For example, a survey of tourists to Oxford on a Tuesday afternoon in April found that most people were attracted to the university buildings and the colleges, and listed the poor weather as their main criticism. Had the survey taken place in another kind of tourist destination, the attractions listed would have been very different. Similarly, had the survey taken place in summer the weather might not have been mentioned as a problem. On the other hand, congestion due to too many tourists might have been mentioned. This example shows that the methods (including the date and time of any survey) produce results that can affect your conclusions and, therefore, your evaluation.

The final part of your evaluation can lead you to suggest future lines of enquiry from the insights you have developed by following the route to geographical enquiry in your investigation.

Coursework assessment

Before you begin your coursework enquiry you should be aware of how your coursework will be assessed. The mark scheme for the assessment of coursework is in the syllabus document. This is in the form of a matrix, which shows what you need to do to meet each of the five assessment criteria.

Activities

- 1 What is the difference between analysis and interpretation?
 - 2 Write an analysis and interpretation of the isoline pattern shown in [Figure 11.9 \(page 297\)](#).
 - 3 What might you include in the evaluation of your coursework?
-

Case studies

Analysing sand dunes



▲ **Figure 11.44** Sand dunes at Studland, Dorset, UK

Sand dunes provide an interesting and manageable ecosystem for study at this level. This is because significant changes can be identified over a relatively small area. In a sand dune system, the most recently formed dunes are by the sea. The dunes become older with distance inland. Sand dunes form a series of ridges with intervening 'slacks' between them.

A useful starting point is to survey the morphology (size and shape) of the dunes. [Figure 11.45a \(page 322\)](#) shows how measurements can be taken across a sand dune ecosystem using a tape measure and a clinometer. The transect line should be at right-angles to the coast. The first ranging pole is carefully placed where there is a distinct break in slope from the back of the beach, marking the beginning of the sand dunes. The second ranging pole is placed at the next break of slope. The angle of slope is read from the clinometer. This process is repeated for each break of slope. With about four people working as a team, all the measurements required to draw a cross-section of the dunes ([Figure 11.45b](#)) can be taken in a couple of hours. If a larger group of people is available a number of transects could be taken across the sand dunes. Transects could be compared and any differences discussed.

As the dune survey proceeds, other measurements can also be taken. At regular locations across the sand

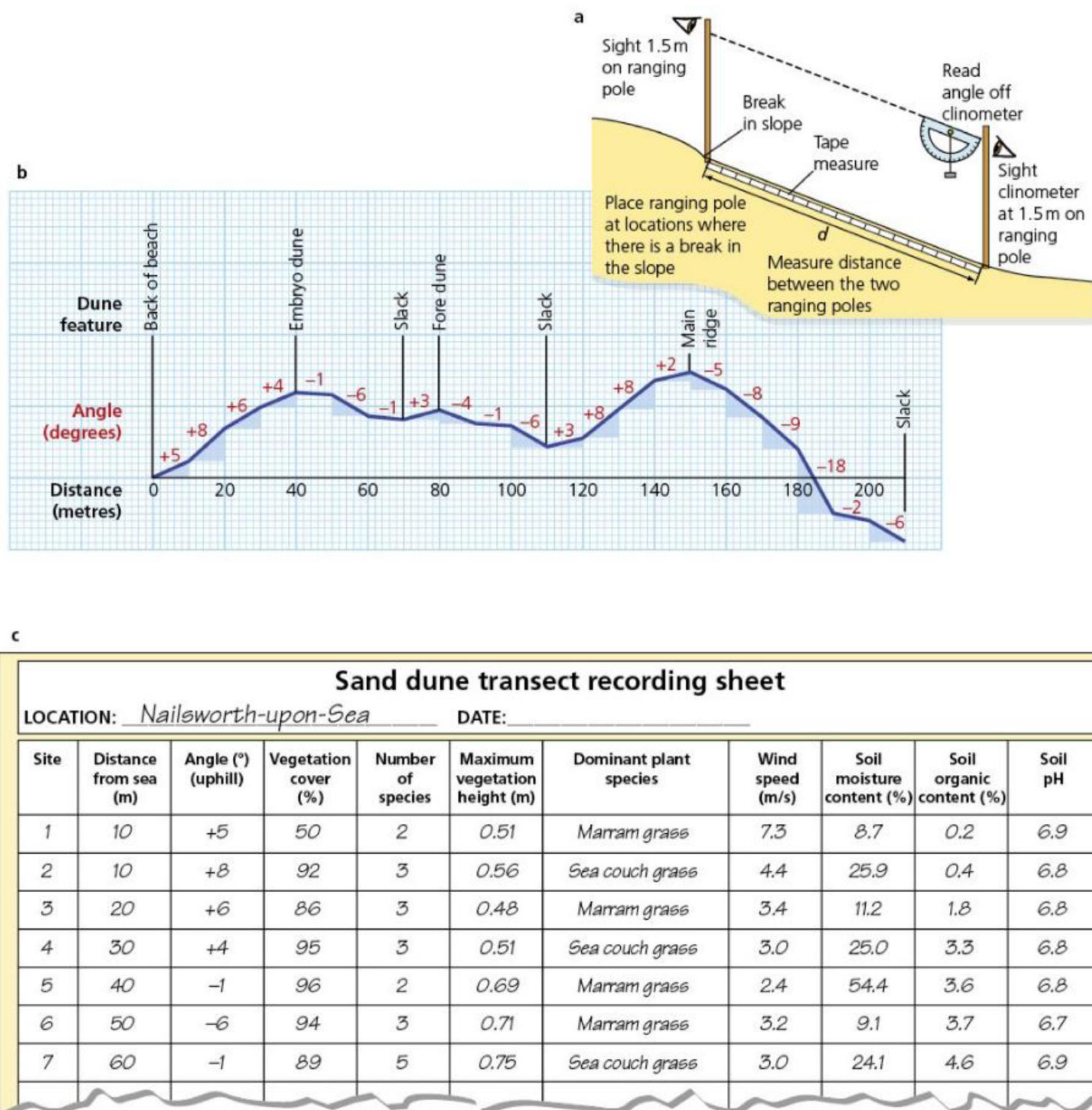
dune system the following can be measured:

- » vegetation cover, with the dominant plant species noted
- » maximum height of vegetation
- » wind speed
- » soil moisture content
- » soil organic content
- » soil pH.

Figure 11.45c provides an example of a recording sheet that could be used for such a survey. While the first few readings might take a little time, once you become familiar with what is required the process should speed up considerably.

All these measurements can be used to test the standard theories about sand dunes presented in textbooks, which can be set out in a series of hypotheses to be tested:

- » Vegetation density increases with distance inland.
- » The number of species increases with distance inland.
- » The height of vegetation increases with distance inland.
- » Organic content of soil increases with distance inland.
- » Soil pH decreases with distance inland.
- » Wind speed decreases with distance inland.

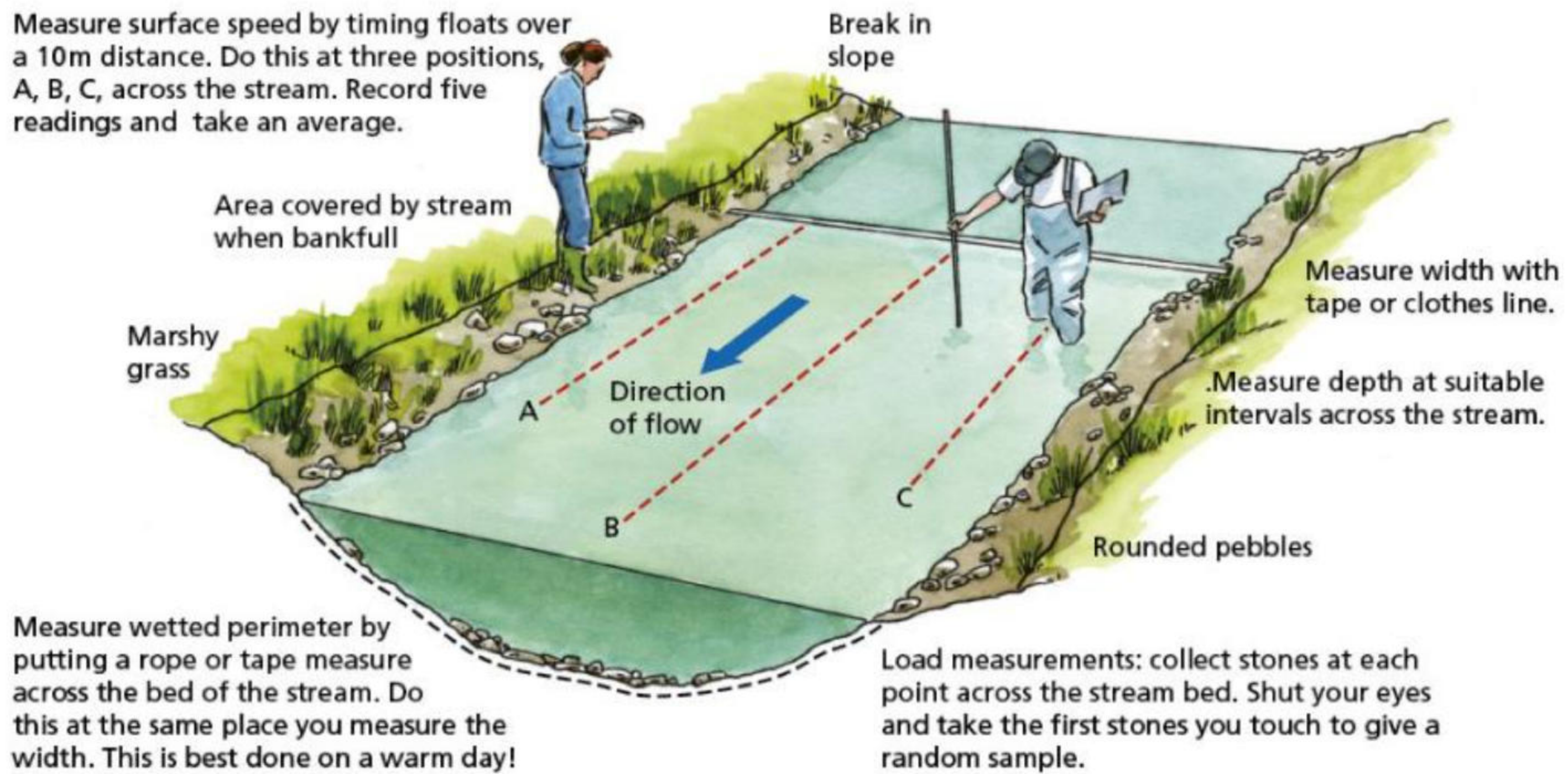


▲ **Figure 11.45** Conducting a survey of a sand dune ecosystem

Investigating rivers

Streams and small rivers are a popular focus for geographical investigation because most schools will not be too far from a suitable example. Figure 11.46 shows some of the measurements that can be taken at

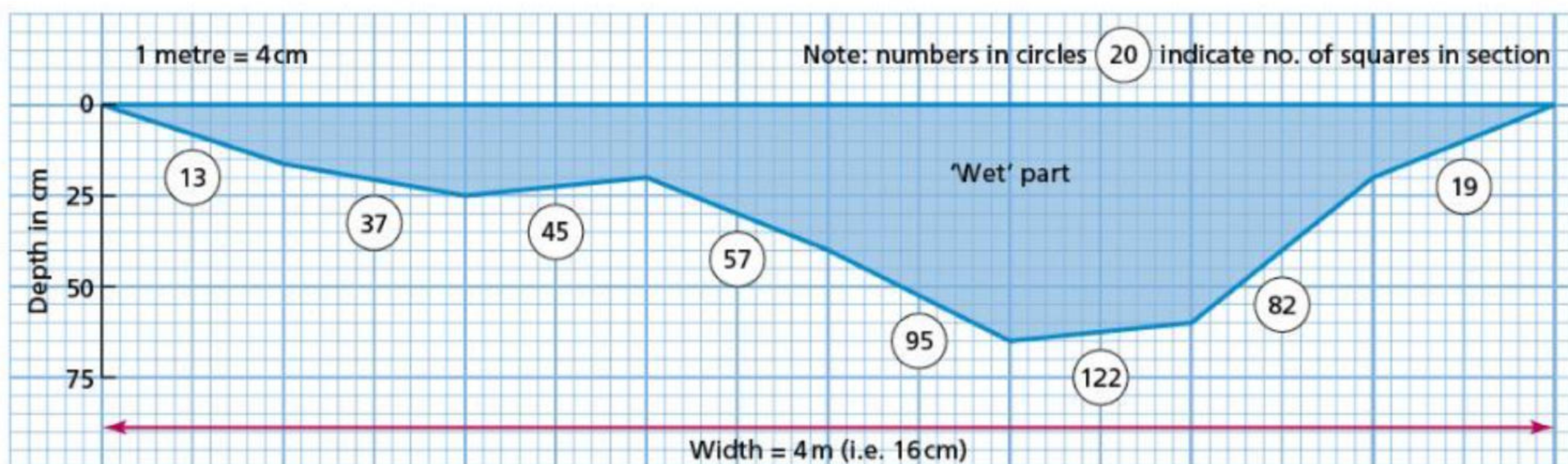
various locations along the course of a river. For safety reasons it is best to avoid working in streams above the height of your knees.



▲ **Figure 11.46** Taking river measurements

For most river studies you will want to produce a cross-section of the river channel. The method is as follows:

- » Use a tape measure to assess the channel width. This should be done at right-angles to the course of the river. If you want to produce a cross-section of the river when discharge is at its highest, you should look for evidence of the highest point the water reaches on each bank. This will give the bankfull width.
- » Channel depth should be measured at regular intervals across the river using a metre stick or ranging pole. Every 20 cm or 30 cm should provide an adequate sampling interval.
- » A cross-section can then be drawn using graph paper ([Figure 11.47](#)). As with all cross-sections, careful choice of scale is important.
- » Use a stopwatch or timer to time how long it takes an orange to travel 10 m in the stream. Repeat at least three times.



Area = 470 squares. This number has to be converted, according to the scale. The scale 4 cm = 1 m means that each 2-millimetre square on the graph paper represents 0.0025 m² in reality ($1/400$). If the scale had been 1 cm = 1 m then the scaling factor would be $1/100 = 0.01$ m². In this example $470 \times 0.0025 = 1.175$ m².

▲ **Figure 11.47** A river cross section

Although various types of float can be used to measure river velocity, it is best to use a flow meter. The impeller (screw device) is pointed upstream at the same points across the river used to calculate the depth intervals. You will be able to see how velocity varies with distance from the banks and how velocity varies with depth. You could also calculate the mean flow rate for this stage of the river.

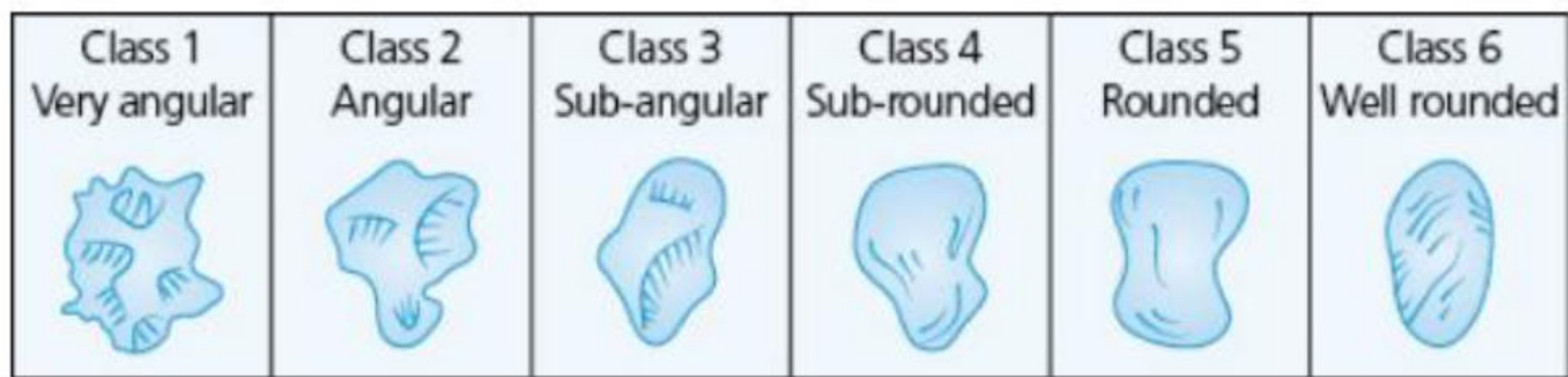
The discharge of the river can be calculated by multiplying the velocity by the cross-sectional area ([Figure 11.47](#)). The gradient of a river can be measured using ranging poles and a clinometer, in the same way that sand dune measurements were carried out in the previous example.

Bedload measurements can also be taken to assess the impact of attrition with increasing distance

downstream. Ensure that the samples of bedload are selected randomly by a ranging pole or metre stick at intervals across the river. Collect the stones that are touching the pole or stick. Measure the long axis, shape and radius of curvature of each stone.

As in the sand dune case study, a series of hypotheses based on textbook theory can be set up to be tested.

Use Power’s scale of roundness ([Figure 11.48](#)) to see whether pebble shape becomes rounder downstream.



▲ **Figure 11.48** Power's scale of roundness